

Title

Unveiling Trends in International Soccer: A Data-Driven Analysis of Match Outcomes

Abstract

Abstract This study explores the evolving trends in international football, focusing on two key aspects: home advantage and goal-scoring patterns. Using a comprehensive dataset of international soccer matches, we examine whether the location of a match correlates with the likelihood of the home team winning. Statistical analyses are employed to quantify the extent of home advantage and uncover patterns in team performance across different eras. Additionally, we investigate the role of penalty kicks in determining match results, analyzing whether having a penalty kick significantly increases the chances of victory. By leveraging advanced statistical techniques, including hypothesis testing and trend analysis, this research provides a deeper understanding of how these factors influence outcomes in international football. The findings offer valuable insights for coaches, analysts, and fans, shedding light on the dynamics that shape the beautiful game on a global stage.

For this research, the following datasets from the "International football results from 1872 to 2024" collection are used. This collection of datasets were obtained from Kaggle:

<https://www.kaggle.com/datasets/martj42/international-football-results-from-1872-to-2017/data>

1. results.csv: Contains match results and metadata, such as the date, teams involved, scores, tournament type, and venue information.
2. shootouts.csv: Includes details about penalty shootouts, such as winners and the sequence of shooters.
3. goalscorers.csv: Offers information on goal scorers, including whether goals were penalties or own goals.

These datasets cover historical international matches, including match results, player-level contributions (such as goals scored in a match) and additional game statistics such as those coming from a penalty shootout.

Introduction

Soccer, often referred to as the beautiful game, is not only the world's most popular sport but also a fertile ground for statistical and predictive analysis. Research in this domain has underscored the significance of various factors influencing match outcomes, such as team form, home-field advantage, and the application of advanced machine learning models. Constantinou, Fenton, and Neil (2012) demonstrated that integrating recent performance data and match context—particularly the home or away status of teams—significantly

improved predictive accuracy. Similarly, Tax and Joutstra (2015) explored multiple machine learning algorithms, finding that random forests excelled in capturing complex interactions among features, such as recent team form and defensive strength. These studies underscore the growing potential of combining team statistics with sophisticated analytical techniques to unravel the intricacies of soccer match outcomes.

Building upon these foundations, this study investigates emerging trends in international football by examining pivotal questions: (1) How strongly does the location of a match correlate with home team success? (2) Does the presence of a penalty kick in a match substantially increase the likelihood of a win? By analyzing a comprehensive dataset of international soccer matches, this project aims to shed light on these aspects, leveraging both traditional statistical methods and modern data-driven approaches. Through this work, we seek to contribute to the understanding of soccer dynamics while providing actionable insights for analysts, strategists, and fans alike.

All this data, together, allows us to carry out a multidimensional exploration of trends, team performance and scoring behaviors. The fact that the data spans matches from 1872 to 2024, it offers us with insights into nearly 150 years of international football history. This increases the reliability of our analysis, allowing for the analysis of long term trends, including changes in football strategies, global participation, and team performance.

The dataset is publicly available on Kaggle and can be used for research purposes. Here is the license listed on the Kaggle website:

<https://creativecommons.org/publicdomain/zero/1.0/>

1. Dataset Structure

```
In [85]: # Import necessary Libraries
import pandas as pd
```

```
In [86]: # Load your dataset
results_df = pd.read_csv('results.csv')
shootouts_df = pd.read_csv('shootouts.csv')
goalscorers_df = pd.read_csv('goalscorers.csv')
```

```
In [87]: # Basic dataset information
print("Results Dataset Info:")
results_df.info()
print()
print("Shootouts Dataset Info:")
shootouts_df.info()
print()
print("Goalscorers Dataset Info:")
goalscorers_df.info()
```

Results Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 47777 entries, 0 to 47776
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   date        47777 non-null  object
1   home_team   47777 non-null  object
2   away_team   47777 non-null  object
3   home_score  47777 non-null  int64
4   away_score  47777 non-null  int64
5   tournament  47777 non-null  object
6   city        47777 non-null  object
7   country     47777 non-null  object
8   neutral     47777 non-null  bool
dtypes: bool(1), int64(2), object(6)
memory usage: 3.0+ MB
```

Shootouts Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 645 entries, 0 to 644
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   date        645 non-null    object
1   home_team   645 non-null    object
2   away_team   645 non-null    object
3   winner      645 non-null    object
4   first_shooter 230 non-null    object
dtypes: object(5)
memory usage: 25.3+ KB
```

Goalscorers Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 44362 entries, 0 to 44361
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   date        44362 non-null  object
1   home_team   44362 non-null  object
2   away_team   44362 non-null  object
3   team        44362 non-null  object
4   scorer      44313 non-null  object
5   minute      44103 non-null  float64
6   own_goal    44362 non-null  bool
7   penalty     44362 non-null  bool
dtypes: bool(2), float64(1), object(5)
memory usage: 2.1+ MB
```

```
In [88]: print("Results Dataset Shape:")
print(results_df.shape)
print("\nShootouts Dataset Shape:")
print(shootouts_df.shape)
print("\nGoalscorers Dataset Shape:")
print(goalscorers_df.shape)
```

Results Dataset Shape:
(47777, 9)

Shootouts Dataset Shape:
(645, 5)

Goalscorers Dataset Shape:
(44362, 8)

```
In [89]: print("Results Dataset Column Names:")
print(results_df.columns)
print("\nShootouts Dataset Column Names:")
print(shootouts_df.columns)
print("\nGoalscorers Dataset Column Names:")
print(goalscorers_df.columns)
```

Results Dataset Column Names:
Index(['date', 'home_team', 'away_team', 'home_score', 'away_score',
 'tournament', 'city', 'country', 'neutral'],
 dtype='object')

Shootouts Dataset Column Names:
Index(['date', 'home_team', 'away_team', 'winner', 'first_shooter'], dtype='object')

Goalscorers Dataset Column Names:
Index(['date', 'home_team', 'away_team', 'team', 'scorer', 'minute',
 'own_goal', 'penalty'],
 dtype='object')

```
In [90]: print("Results Dataset Data Types:")
print(results_df.dtypes)
print("\nShootouts Dataset Data Types:")
print(shootouts_df.dtypes)
print("\nGoalscorers Dataset Data Types:")
print(goalscorers_df.dtypes)
```

Results Dataset Data Types:

```

date          object
home_team     object
away_team     object
home_score    int64
away_score    int64
tournament    object
city          object
country       object
neutral       bool
dtype: object

```

Shootouts Dataset Data Types:

```

date          object
home_team     object
away_team     object
winner        object
first_shooter object
dtype: object

```

Goalscorers Dataset Data Types:

```

date          object
home_team     object
away_team     object
team          object
scorer        object
minute        float64
own_goal      bool
penalty       bool
dtype: object

```

```

In [91]: # List of major tournaments to filter
major_tournaments = [
    "FIFA World Cup",
    "UEFA European Championship",
    "Copa América",
    "AFC Asian Cup",
    "CONCACAF Gold Cup",
    "Africa Cup of Nations",
    "Oceania Nations Cup"
]

# 1. Filter 'results.csv'
filtered_results = results_df[results_df['tournament'].isin(major_tournaments)]

print("Filtered Results Dataset Shape:", filtered_results.shape)
print("Preview of Filtered Results Dataset:")
print(filtered_results.head())

# 2. Filter 'shootouts.csv'
# Join 'shootouts' with 'results' to get the tournament information
filtered_shootouts = shootouts_df.merge(
    results_df[['date', 'home_team', 'away_team', 'tournament']],
    on=['date', 'home_team', 'away_team'],
    how='inner'
)

```

```
filtered_shootouts = filtered_shootouts[filtered_shootouts['tournament'].isin(major)]

print()
print("Filtered Shootouts Dataset Shape:", filtered_shootouts.shape)
print("Preview of Filtered Shootouts Dataset:")
print(filtered_shootouts.head())

# 3. Filter 'goalscorers.csv'
# Join 'goalscorers' with 'results' to get the tournament information
filtered_goalscorers = goalscorers_df.merge(
    results_df[['date', 'home_team', 'away_team', 'tournament']],
    on=['date', 'home_team', 'away_team'],
    how='inner'
)
filtered_goalscorers = filtered_goalscorers[filtered_goalscorers['tournament'].isin(major)]

print()
print("Filtered Goalscorers Dataset Shape:", filtered_goalscorers.shape)
print("Preview of Filtered Goalscorers Dataset:")
print(filtered_goalscorers.head())
```

Filtered Results Dataset Shape: (2399, 9)

Preview of Filtered Results Dataset:

	date	home_team	away_team	home_score	away_score	tournament	\
462	1916-07-02	Chile	Uruguay	0	4	Copa América	
464	1916-07-06	Argentina	Chile	6	1	Copa América	
465	1916-07-08	Brazil	Chile	1	1	Copa América	
466	1916-07-10	Argentina	Brazil	1	1	Copa América	
468	1916-07-12	Brazil	Uruguay	1	2	Copa América	

	city	country	neutral
462	Buenos Aires	Argentina	True
464	Buenos Aires	Argentina	False
465	Buenos Aires	Argentina	True
466	Buenos Aires	Argentina	False
468	Buenos Aires	Argentina	True

Filtered Shootouts Dataset Shape: (96, 6)

Preview of Filtered Shootouts Dataset:

	date	home_team	away_team	winner	first_shooter	\
2	1972-05-07	South Korea	Iraq	Iraq	NaN	
3	1972-05-17	Thailand	South Korea	South Korea	NaN	
4	1972-05-19	Thailand	Cambodia	Thailand	NaN	
53	1982-07-08	Germany	France	Germany	France	
76	1984-12-13	Saudi Arabia	Iran	Saudi Arabia	NaN	

	tournament
2	AFC Asian Cup
3	AFC Asian Cup
4	AFC Asian Cup
53	FIFA World Cup
76	AFC Asian Cup

Filtered Goalscorers Dataset Shape: (7160, 9)

Preview of Filtered Goalscorers Dataset:

	date	home_team	away_team	team	scorer	minute	\
0	1916-07-02	Chile	Uruguay	Uruguay	José Piendibene	44.0	
1	1916-07-02	Chile	Uruguay	Uruguay	Isabelino Gradín	55.0	
2	1916-07-02	Chile	Uruguay	Uruguay	Isabelino Gradín	70.0	
3	1916-07-02	Chile	Uruguay	Uruguay	José Piendibene	75.0	
4	1916-07-06	Argentina	Chile	Argentina	Alberto Ohaco	2.0	

	own_goal	penalty	tournament
0	False	False	Copa América
1	False	False	Copa América
2	False	False	Copa América
3	False	False	Copa América
4	False	False	Copa América

Dataset Size and Scope

Complete Results Dataset:

- Observation: 47,777
- Variables: 9

Complete Shootouts Dataset:

- Observation: 645
- Variables: 5

Complete Goalscorers Dataset:

- Observation: 44,362
- Variables: 8

After doing some filtering, we are considering only carrying out the analysis of the data from major tournaments. These tournaments consist of the following:

- FiFa World Cup
- UEFA European Championship
- Copa América
- AFC Asian Cup
- CONCACAF Gold Cup
- Africa Cup of Nations
- Oceania Nations Cup

These new subsets gives us the following information:

Results Subset:

- Observation: 2,399
- Variables: 9

Shootouts Subset:

- Observation: 96
- Variables: 4 (We are planning to only use 4 of the variables)

Goalscorers Subset:

- Observation: 7,160
- Variables: 6 (We are planning to only use 6 of the variables)

Variable Details

Results.csv:

These are the variables we are looking to use:

Variable Name	Description	Type
date	Date of the match	Continuous
home_team	The name of the home team	Categorical

Variable Name	Description	Type
away_team	The name of the away team	Categorical
home_score	Full-time home team score including extra time, not including penalty-shootouts	Continuous
away_score	Full-time away team score including extra time, not including penalty-shootouts	Continuous
tournament	The name of the tournament	Categorical
city	The name of the city where the match was played	Categorical
country	The name of the country where the match was played	Categorical
neutral	Boolean indicating if the match was played at a neutral venue	Categorical

Shootouts.csv:

These are the variables we are looking to use:

Variable Name	Description	Type
date	Date of the match	Continuous
home_team	The name of the home team	Categorical
away_team	The name of the away team	Categorical
winner	Winner of the penalty shootout	Categorical
first_shooter	The team that went first in the shootout	Categorical

Goalscorers.csv:

These are the variables we are looking to use:

Variable Name	Description	Type
date	Date of the match	Continuous
home_team	The name of the home team	Categorical
away_team	The name of the away team	Categorical
team	Name of the team scoring the goal	Categorical
scorer	Name of the player scoring the goal	Categorical
own_goal	Boolean indicating whether the goal was an own-goal	Categorical
penalty	Boolean indicating whether the goal was a penalty	Categorical

The first 5 Rows of the dataset is as follows:

```
In [92]: res_so_df = pd.merge(
    filtered_results,
    filtered_shootouts,
    on=['date', 'home_team', 'away_team'],
    how='left' # Use 'left' join to retain all rows from filtered_results
)

df = pd.merge(
    res_so_df,
    filtered_goalscorers,
    on=['date', 'home_team', 'away_team'],
    how='left' # Use 'left' join to retain all rows from results_shootouts_merged
)
df.head()
```

Out[92]:

	date	home_team	away_team	home_score	away_score	tournament_x	city	coun
0	1916-07-02	Chile	Uruguay	0	4	Copa América	Buenos Aires	Argent
1	1916-07-02	Chile	Uruguay	0	4	Copa América	Buenos Aires	Argent
2	1916-07-02	Chile	Uruguay	0	4	Copa América	Buenos Aires	Argent
3	1916-07-02	Chile	Uruguay	0	4	Copa América	Buenos Aires	Argent
4	1916-07-06	Argentina	Chile	6	1	Copa América	Buenos Aires	Argent

```
In [93]: # Display last few records
print("\nLast 5 records:")
df.tail()
```

Last 5 records:

Out[93]:

	date	home_team	away_team	home_score	away_score	tournament_x	city	c
7339	2024-07-13	Canada	Uruguay	2	2	Copa América	Charlotte	
7340	2024-07-13	Canada	Uruguay	2	2	Copa América	Charlotte	
7341	2024-07-13	Canada	Uruguay	2	2	Copa América	Charlotte	
7342	2024-07-13	Canada	Uruguay	2	2	Copa América	Charlotte	
7343	2024-07-14	Argentina	Colombia	1	0	Copa América	Miami Gardens	

```
In [94]: # Display basic summary statistics
print("\nSummary Statistics:")
df.describe()
```

Summary Statistics:

```
Out[94]:
```

	home_score	away_score	minute
count	7344.000000	7344.000000	7038.000000
mean	2.584423	1.737881	50.135408
std	2.327288	1.705724	27.073477
min	0.000000	0.000000	1.000000
25%	1.000000	1.000000	27.000000
50%	2.000000	1.000000	51.000000
75%	4.000000	2.000000	73.000000
max	17.000000	12.000000	122.000000

```
In [95]: # Missing Values in a variable
print("Number of missing values in [tournament]:")
print(df['tournament'].isna().sum())
```

Number of missing values in [tournament]:
184

```
In [96]: # To see unique values
print("Unique values in [first_shooter]:")
unique_array = df['first_shooter'].unique()
print(len(unique_array))
print(unique_array)
```

Unique values in [first_shooter]:
40
[nan 'France' 'Brazil' 'Germany' 'Spain' 'South Korea' 'Romania'
'Argentina' 'Italy' 'England' 'Colombia' 'Mexico' 'Sweden'
'United States' 'Uruguay' 'Iran' 'Kuwait' 'Saudi Arabia' 'Ecuador'
'Republic of Ireland' 'Bahrain' 'Japan' 'China PR' 'Ukraine' 'Portugal'
'Australia' 'Paraguay' 'Costa Rica' 'Netherlands' 'Chile' 'New Zealand'
'Denmark' 'Russia' 'Vietnam' 'Morocco' 'Croatia' 'Tajikistan'
'Uzbekistan' 'Venezuela' 'Canada']

```
In [97]: # To see frequencies (value counts)
print("\nValue counts in [scorer]:")
df['scorer'].value_counts()
```

Value counts in [scorer]:

```
Out[97]: scorer
Lionel Messi      27
Ronaldo           25
Gabriel Batistuta 23
Ademir de Menezes 22
Pelé              20
..
Sávio            1
Omar Alderete     1
Jose Tognola       1
Manoel Alencar Monte 1
José Durand Laguna 1
Name: count, Length: 3231, dtype: int64
```

```
In [98]: # To see frequencies with percentages
print("\nValue counts with percentages:")
df['scorer'].value_counts(normalize=True)
```

Value counts with percentages:

```
Out[98]: scorer
Lionel Messi      0.003797
Ronaldo           0.003516
Gabriel Batistuta 0.003234
Ademir de Menezes 0.003094
Pelé              0.002813
...
Sávio            0.000141
Omar Alderete     0.000141
Jose Tognola       0.000141
Manoel Alencar Monte 0.000141
José Durand Laguna 0.000141
Name: proportion, Length: 3231, dtype: float64
```

From the snippet of the data, we are able to understand a little bit of the dataset. The summary statistics shows that there may be some potential errors as well as outliers in the data. For example, the number of goals made between home and away points are in ratio of 1:1. The chance of this happening is so small but is not impossible. Furthermore, the outliers such as large number of points can mess up the study and may need to be readjusted/removed. Other than mentioned and found, we believe that this dataset qualifies to be used for this predictive analysis project. This is also supported from the non-existing NA values.

Dataset Cleaning

Before we begin the Dataset Cleaning, we have taken the feedback we received into account and converted our code to use merge instead of concat to combin our three datasets into one.

```
In [99]: # Merging the datasets on common keys: 'date', 'home_team', and 'away_team'
```

```

# Merge filtered_results and filtered_shootouts
results_shootouts_merged = pd.merge(
    filtered_results,
    filtered_shootouts,
    on=['date', 'home_team', 'away_team'],
    how='left' # Use 'left' join to retain all rows from filtered_results
)

# Merge the above result with filtered_goalscorers
final_merged_data = pd.merge(
    results_shootouts_merged,
    filtered_goalscorers,
    on=['date', 'home_team', 'away_team'],
    how='left' # Use 'left' join to retain all rows from results_shootouts_merged
)

# Display the shape and sample of the final merged dataset
print("Final Merged Dataset Shape:", final_merged_data.shape)
print(final_merged_data.head())

```

Final Merged Dataset Shape: (7344, 18)

	date	home_team	away_team	home_score	away_score	tournament_x \
0	1916-07-02	Chile	Uruguay	0	4	Copa América
1	1916-07-02	Chile	Uruguay	0	4	Copa América
2	1916-07-02	Chile	Uruguay	0	4	Copa América
3	1916-07-02	Chile	Uruguay	0	4	Copa América
4	1916-07-06	Argentina	Chile	6	1	Copa América

	city	country	neutral	winner	first_shooter	tournament_y \
0	Buenos Aires	Argentina	True	NaN	NaN	NaN
1	Buenos Aires	Argentina	True	NaN	NaN	NaN
2	Buenos Aires	Argentina	True	NaN	NaN	NaN
3	Buenos Aires	Argentina	True	NaN	NaN	NaN
4	Buenos Aires	Argentina	False	NaN	NaN	NaN

	team	scorer	minute	own_goal	penalty	tournament
0	Uruguay	José Piendibene	44.0	False	False	Copa América
1	Uruguay	Isabelino Gradín	55.0	False	False	Copa América
2	Uruguay	Isabelino Gradín	70.0	False	False	Copa América
3	Uruguay	José Piendibene	75.0	False	False	Copa América
4	Argentina	Alberto Ohaco	2.0	False	False	Copa América

Part 1: Missing Value Analysis

1. Missing Value Assessment

In [100...

```

import matplotlib.pyplot as plt
import seaborn as sns

final_merged_data = final_merged_data.drop(columns=['tournament_y'])
final_merged_data = final_merged_data.drop(columns=['tournament_x'])

# Analyze missing values in the dataset
missing_values = final_merged_data.isnull().sum()
missing_percent = (missing_values / len(final_merged_data)) * 100

```

```

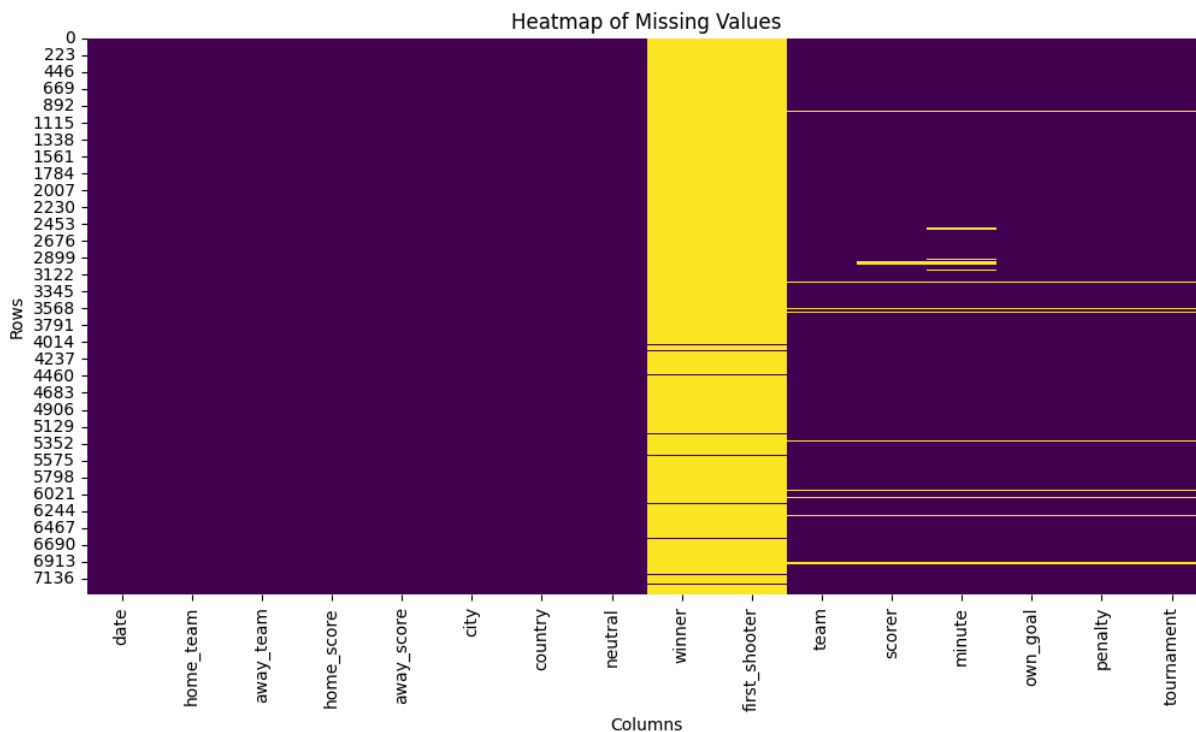
# Create a summary DataFrame for missing values
missing_summary = pd.DataFrame({
    'Column': missing_values.index,
    'Missing Values': missing_values.values,
    'Percentage Missing': missing_percent.values
}).sort_values(by='Missing Values', ascending=False)

# Display the summary table
print(missing_summary)

# Visualize missing data with a heatmap
plt.figure(figsize=(12, 6))
sns.heatmap(final_merged_data.isnull(), cbar=False, cmap="viridis")
plt.title("Heatmap of Missing Values")
plt.xlabel("Columns")
plt.ylabel("Rows")
plt.show()

```

	Column	Missing Values	Percentage Missing
9	first_shooter	7157	97.453704
8	winner	7143	97.263072
12	minute	306	4.166667
11	scorer	233	3.172658
15	tournament	184	2.505447
14	penalty	184	2.505447
13	own_goal	184	2.505447
10	team	184	2.505447
7	neutral	0	0.000000
6	country	0	0.000000
5	city	0	0.000000
4	away_score	0	0.000000
3	home_score	0	0.000000
2	away_team	0	0.000000
1	home_team	0	0.000000
0	date	0	0.000000



Observations from looking at missing values:

- winner (of a penalty shootout), first_shooter, tournament_y:
Over 97% missing values, likely because shootout data is only relevant for a small subset of matches that went to penalties.
- team, scorer, minute:
A small percentage of missing values, possibly tied to matches with no goals scored.
- own_goal, penalty, tournament:
Same level of missingness as team, suggesting these variables are interrelated.
- The scattered missingness in goal-related variables suggests these may be dependent on the outcome of matches

1. winner and first_shooter (Penalty Shootout Variables)

- These variables are only relevant for matches that went to a penalty shootout. For matches that did not go to penalties, they have no value.
- MAR - missingness depends on whether the match included a penalty shootout

2. team, scorer, and minute (Goal-Related Variables)

- These variables are tied to matches where goals were scored. Matches with no goals have no corresponding entries for these variables.
- It could also be that the data was not recorded or lost.
- MCAR.
- MAR - missigness is conditional on match outcomes, whether goals were scored.

3. own_goal, penalty (Goal Details)

- These variables depend on whether goals were scored, similar to team and scorer.
- MAR - missingness is conditional of goals.

None of the variables with missing data appear to be MCAR

2. Missing Value Treatment

winner and first_shooter can be dropped. These are irrelevant to our research question as trends in penalty shootout outcomes do not directly relate to home advantage, goal-scoring, or team strength.

```
In [101... final_merged_data = final_merged_data.drop(columns=['winner'])
final_merged_data = final_merged_data.drop(columns=['first_shooter'])
```

team, scorer, and, minute can be handled the following way:

- Handle Matches with No Goals: For rows where both home_score and away_score are 0, fill missing values in team with 'No Goals'. Similarly, for scorer and minute, missing values in these matches indicate no goals, so fill them with appropriate placeholders.
- Listwise Deletion for Remaining Missing Values: For rows where missing data is truly random and accounts for less than 5%, drop them to maintain data quality.

own_goal and penalty will be handled the same way.

```
In [102... # Identify matches with no goals (both home and away scores are 0)
no_goals_mask = (final_merged_data['home_score'] == 0) & (final_merged_data['away_s

# Fill missing values for matches with no goals
final_merged_data.loc[no_goals_mask, 'team'] = 'No Goals'
final_merged_data.loc[no_goals_mask, 'scorer'] = 'No Goals'
final_merged_data.loc[no_goals_mask, 'minute'] = 0 # Assuming 0 for 'minute' to in
final_merged_data.loc[no_goals_mask, 'own_goal'] = 0 # Assuming 0 indicates no own
final_merged_data.loc[no_goals_mask, 'penalty'] = 0 # Assuming 0 indicates no pen

# Perform Listwise Deletion for remaining rows with missing values in goal-related
goal_related_columns = ['team', 'scorer', 'minute', 'own_goal', 'penalty']
final_merged_data = final_merged_data.dropna(subset=goal_related_columns)

# Verify that there are no missing values left in the goal-related columns
print("Remaining Missing Values in Goal-Related Columns:")
print(final_merged_data[goal_related_columns].isnull().sum())
```

Remaining Missing Values in Goal-Related Columns:

```
team      0
scorer    0
minute    0
own_goal  0
penalty   0
dtype: int64
```


For the tournament column, the missing data (2.51%) is likely due to matches that were not part of a recognized or documented tournament (e.g., friendlies). This could result from unrecorded or lost data (MCAR) or the nature of certain matches (MAR).

We will fill the missing values with placeholder, "Unknown Tournament".

```
In [103... # Fill missing values in 'tournament' with a placeholder
final_merged_data['tournament'] = final_merged_data['tournament'].fillna('Unknown T
print(final_merged_data["tournament"].isnull().sum())
```

0

Part 2: Outlier Detection and Treatment

```
In [104... import seaborn as sns

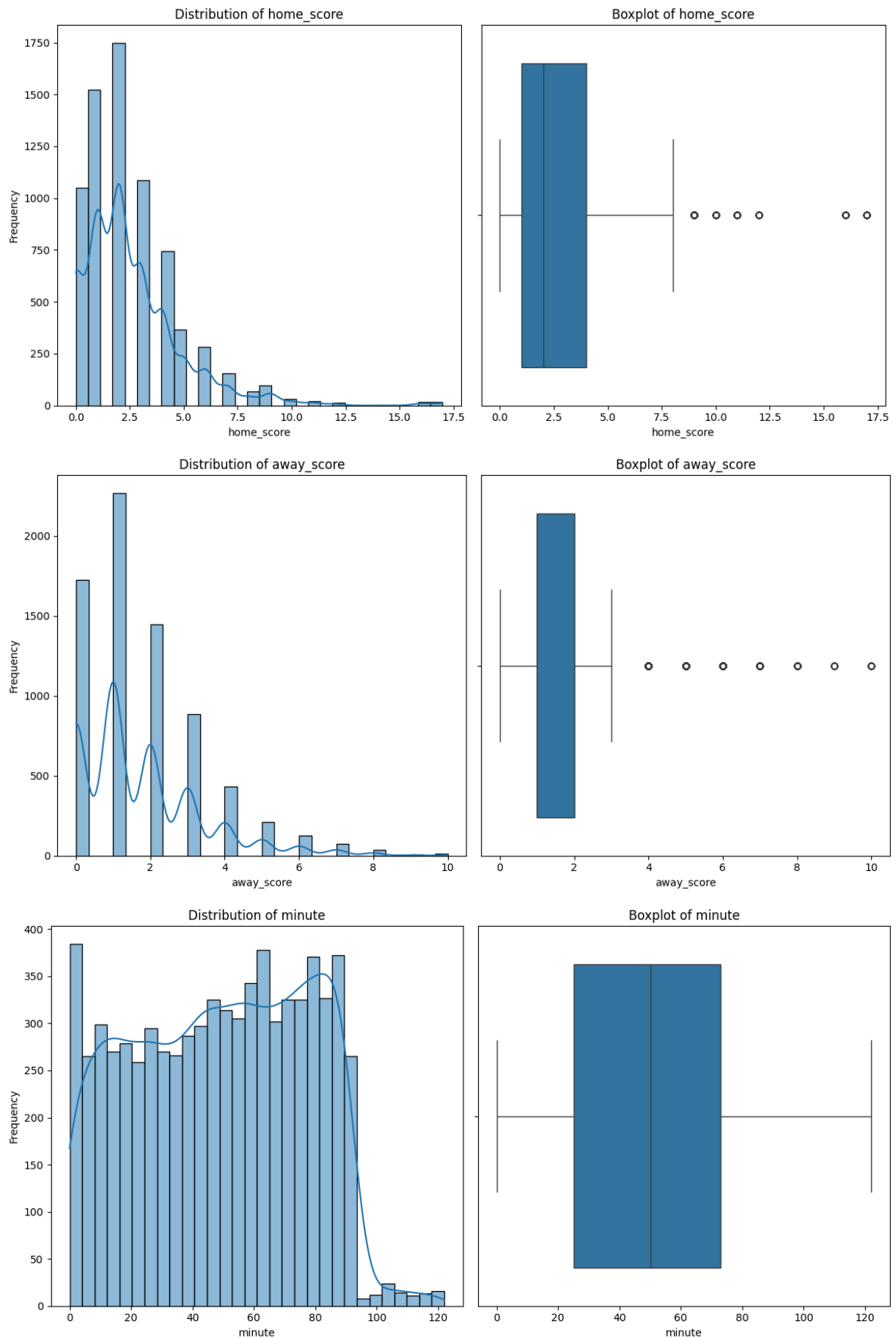
# Numeric columns of interest
numeric_columns = ['home_score', 'away_score', 'minute']

# Plot histograms and boxplots for each numerical variable
for column in numeric_columns:
    plt.figure(figsize=(12, 6))

    # Histogram
    plt.subplot(1, 2, 1)
    sns.histplot(final_merged_data[column], kde=True, bins=30)
    plt.title(f'Distribution of {column}')
    plt.xlabel(column)
    plt.ylabel('Frequency')

    # Boxplot
    plt.subplot(1, 2, 2)
    sns.boxplot(x=final_merged_data[column])
    plt.title(f'Boxplot of {column}')
    plt.xlabel(column)

plt.tight_layout()
plt.show()
```



Based on the visualizations above:

- `home_score` and `away_score`:
Both are highly skewed, with a significant concentration of low scores (close to 0) and a long right tail for higher scores. IQR would be better for outlier detection here because it is more robust for non-normal distributions.
- `minute`:
This appears to be closer to a normal distribution, as the data is spread relatively symmetrically. Z-score would be better because it is more suitable for normally distributed data

In [105...

```

from scipy import stats
import numpy as np

# Function to remove outliers using IQR
def remove_outliers_iqr(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    # Remove outliers
    return df[(df[column] >= lower_bound) & (df[column] <= upper_bound)]

# Function to remove outliers using Z-score
def remove_outliers_zscore(df, column, threshold=3):
    z_scores = stats.zscore(df[column], nan_policy='omit')
    return df[np.abs(z_scores) <= threshold]

# Apply IQR for 'home_score' and 'away_score'
final_merged_data = remove_outliers_iqr(final_merged_data, 'home_score')
final_merged_data = remove_outliers_iqr(final_merged_data, 'away_score')

# Apply Z-score for 'minute'
final_merged_data = remove_outliers_zscore(final_merged_data, 'minute')

# Verify the remaining values in numeric columns
numeric_columns = ['home_score', 'away_score', 'minute']

# Display the summary for numeric columns
for column in numeric_columns:
    print(f"Column: {column}")
    print(f"  Min: {final_merged_data[column].min()}")
    print(f"  Max: {final_merged_data[column].max()}")
    print(f"  Remaining Rows: {len(final_merged_data)}")
    print()

```

Column: home_score
 Min: 0
 Max: 8
 Remaining Rows: 6894

Column: away_score
 Min: 0
 Max: 6
 Remaining Rows: 6894

Column: minute
 Min: 0.0
 Max: 122.0
 Remaining Rows: 6894

Part 3: Variable Correlation Analysis:

```
In [106... # Create a dataframe that removes duplicates of the same games and filter only df['
non_neutral_df = final_merged_data.copy()
non_neutral_df = non_neutral_df[non_neutral_df['neutral'] == False]
non_neutral_df = non_neutral_df.drop_duplicates(subset=['date', 'home_team', 'away_
```

```
In [107... # Label the 'is_winner' column with the home_team value if home_score > away_score
non_neutral_df['is_winner'] = np.where(
    non_neutral_df['home_score'] > non_neutral_df['away_score'],
    non_neutral_df['home_team'], # Assign home_team value
    None # Assign None if condition is not met
)
```

```
In [108... # Drop the non-numeric columns like 'date', 'home_team', 'away_team', etc.
non_neutral_df['is_winner_in_country'] = (non_neutral_df['is_winner'] == non_neutra
numeric_df = non_neutral_df.select_dtypes(include=['number'])

# Calculate the correlation matrix
correlation_matrix = numeric_df.corr()

# Display the correlation with 'is_winner_in_country'
print(correlation_matrix['is_winner_in_country'])
```

```
home_score          0.602196
away_score         -0.522310
minute             0.071150
is_winner_in_country 1.000000
Name: is_winner_in_country, dtype: float64
```

From this correlation result, we can see that winning is correlated with having home_team to win. Although it's not always the case, but the correlation of home team winning with home team scoring is positive. Meanwhile, the away_team are expected to lose when competing not on their own country.

These correlation values are compared between the amount of goals/score teams created with the variable when they're playing as a home team.

Another correlation we're trying to figure out is between having penalty kicks and winning. Does having a penalty means that teams will win?

We will need to revert back our dataset and adjust as necessary.

```
In [109... # First we'll need to figure out which values are goals that made out of a penalty
non_neutral_df = final_merged_data.copy()
non_neutral_df = non_neutral_df.drop_duplicates(subset=['date', 'home_team', 'away_

print("The number of unique matches: ", non_neutral_df.shape[0])
```

The number of unique matches: 2738

```
In [110... # We'll need to find out who won the game from each row by again comparing the scor
non_neutral_df['team_winner'] = np.where(
    non_neutral_df['home_score'] > non_neutral_df['away_score'],
    non_neutral_df['home_team'], # Assign home_team value
    non_neutral_df['away_team'] # Assign None if condition
)

# is_winner will need to be an integer value so we can use it as a correlation vari
non_neutral_df['is_winner'] = np.where(
    non_neutral_df['team_winner'] == non_neutral_df['team'], # Check if team_winne
    1, # Assign 1 if they match
    0 # Assign 0 if they do not match
)

# change the penalty value to 1 and 0
non_neutral_df['penalty'] = non_neutral_df['penalty'].astype(int)

non_neutral_df['is_winner'] = pd.to_numeric(non_neutral_df['is_winner'], errors='co
non_neutral_df['penalty'] = pd.to_numeric(non_neutral_df['penalty'], errors='coerce

penalty_correlation = non_neutral_df['is_winner'].corr(non_neutral_df['penalty'])
print("Correlation of having a penalty and winning the match:", penalty_correlation

non_neutral_df.dropna(subset=['is_winner', 'penalty'], inplace=True)

# Calculate the correlation matrix between 'penalty' and 'is_winner'
corr_matrix = non_neutral_df[['penalty', 'is_winner']].corr()
```

Correlation of having a penalty and winning the match: -0.057883018932532955

From the result above, we can see that having penalty kicks doesn't guarantee a team to win. The correlation value of -0.058 shows very little or almost no correlation between having penalty and winning the game.

```
In [111... final_merged_data.reset_index(drop=True)
```

Out[111...

	date	home_team	away_team	home_score	away_score	city	country	neutral
0	1916-07-02	Chile	Uruguay	0	4	Buenos Aires	Argentina	True
1	1916-07-02	Chile	Uruguay	0	4	Buenos Aires	Argentina	True
2	1916-07-02	Chile	Uruguay	0	4	Buenos Aires	Argentina	True
3	1916-07-02	Chile	Uruguay	0	4	Buenos Aires	Argentina	True
4	1916-07-06	Argentina	Chile	6	1	Buenos Aires	Argentina	False
...	...	...	...	...	...	...	...	...
6889	2024-07-13	Canada	Uruguay	2	2	Charlotte	United States	True
6890	2024-07-13	Canada	Uruguay	2	2	Charlotte	United States	True
6891	2024-07-13	Canada	Uruguay	2	2	Charlotte	United States	True
6892	2024-07-13	Canada	Uruguay	2	2	Charlotte	United States	True
6893	2024-07-14	Argentina	Colombia	1	0	Miami Gardens	United States	True

6894 rows × 14 columns



Part 5: Conclusion

Data Cleaning

- We found over 97% missing values on winner, first_shooter, tournament_y, that might happen due to shootout data is only relevant for a small subset of matches that went to penalties.
- A small percentage of missing values on team, scorer, and minute, that might happen due to no goals scored.
- Missing values on own_goal, penalty, tournament suggests these variables are interrelated.

In cleaning the dataset, we first need to decide and think of which variables are important for answering our research questions. We were able to remove some columns that are irrelevant to our research question. Moreover, we handle missing values for no goal matches

with labels, 'No Goals'. As for the small number of missing values where the missing data is random and accounts for less than 5%, we drop them to maintain data quality.

After the data cleaning, there are 0 missing values in Goal-related columns.

There are several outliers within our dataset and we managed to handle them with IQR and Z-Score. These methods allowed us to now work with values that stay within our range.

There aren't really any duplicates if we're looking at all variables, but some matches are repeated on the list and needed to be cleaned when necessary.

Correlation

Our main questions are,

1. Does location of match correlates with home team winning?
2. Does having a penalty kick in a match guarantees a win?

To answer #1, We managed to adjust the table with variables that may be necessary in calculating the correlation. We first needed to make sure that there are no duplicate matches to reduce redundancy of the data being used. From there we decided to have a new variable that shows whether or not the winning team is playing in their home country. It shows that there is a moderate correlation between playing as a home team and winning the match compared to playing as an away team and winning the match. A positive correlation is shown for home_score when playing in home country as negative correlation happens for the away_score. For this problem, we filtered the neutral variable to only contain False values so that neutral location for both teams are not considered because they would be outliers for current dataset.

Based on our research, having a penalty kick does not correlate to winning a match. Out of 2738 unique matches, we were able to get a correlation coefficient of -0.05788 which is very close to 0, showing no correlation for teams to win when they are granted penalty kicks.

Final Dataset Characteristics

Our final dataset shows a total of 6894 rows of data with 14 variables in total. This matches are repeated as many as the amount of goals in the match due to the fact that scorers and minute values are different.

When comparing unique matches, there are 2342 matches.

We think that our dataset is now suitable for further analysis and modeling. We handled missing values and redundancy, and outliers. It is also a reliable dataset to used for modeling because we have done variable selection and only relevant variables for our research question.

Data Analysis

Part 1: Descriptive Statistics

For these statistics, we will need to remove the duplicate matches

```
In [112... # Final and cleaned dataset: final_merged_data
no_duplicates_final = final_merged_data.copy()
no_duplicates_final = non_neutral_df.drop_duplicates(subset=['date', 'home_team', '

print("Descriptive Statistics")
no_duplicates_final.describe()
```

Descriptive Statistics

	home_score	away_score	minute	penalty	is_winner
count	2342.000000	2342.000000	2342.000000	2342.000000	2342.000000
mean	1.666097	1.201110	28.029889	0.065756	0.746371
std	1.543460	1.248027	24.707930	0.247908	0.435181
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	0.000000	8.000000	0.000000	0.000000
50%	1.000000	1.000000	21.000000	0.000000	1.000000
75%	2.000000	2.000000	43.000000	0.000000	1.000000
max	8.000000	6.000000	122.000000	1.000000	1.000000

Given the final result of the dataset above shows a cleaned data; equal amount of home_score, away_score and minutes in the table. We can see the dispersion of each of the variable as well.

Below is a more detailed result of the mean, median, and mode of the result.

```
In [113... print("home_score mean value:", no_duplicates_final['home_score'].mean())
print("home_score median value:", no_duplicates_final['home_score'].median())
print("home_score mode value:", no_duplicates_final['home_score'].mode().iloc[0])
print()
print("away_score mean value:", no_duplicates_final['away_score'].mean())
print("away_score median value:", no_duplicates_final['away_score'].median())
print("away_score mode value:", no_duplicates_final['away_score'].mode().iloc[0])
print()
print("minute goal made mean value:", no_duplicates_final['minute'].mean())
print("minute goal made median value:", no_duplicates_final['minute'].median())
print("minute goal made mode value:", no_duplicates_final['minute'].mode().iloc[0])
```



```
home_score mean value: 1.6660973526900085
```

```
home_score median value: 1.0
```

```
home_score mode value: 1
```

```
away_score mean value: 1.2011101622544833
```

```
away_score median value: 1.0
```

```
away_score mode value: 0
```

```
minute goal made mean value: 28.02988898377455
```

```
minute goal made median value: 21.0
```

```
minute goal made mode value: 0.0
```

From the descriptive statistic above, we can see that home_score has higher mean value compared to away_score. This supports our hypothesis of playing at home has a higher chance of winning the game by scoring more goals.

But, we also need to remember that this does not conclude our research due to the situation of neutral locations for these matches.

2. Categorical Analysis

For categorical variables:

- Calculate and interpret frequency distributions
- If applicable and needed, create cross-tabulations for relevant variable pairs
 - Purpose: Displays the interaction between two categorical variables, helping to uncover relationships between categories.
 - Example: Analyzing a cross-tabulation of gender and product purchase might show that females are more likely to purchase a particular product than males, providing valuable input for marketing strategies.
- If applicable and needed, conduct Chi-square tests of independence.
 - Purpose: Tests whether there is a statistically significant association between two categorical variables.
 - Example: For instance, a Chi-square test might reveal that gender significantly impacts customer feedback preferences (p-value < 0.05), offering key insights for customer segmentation strategies.

```
In [114... home_score_freq = no_duplicates_final['home_score'].value_counts()
away_score_freq = no_duplicates_final['away_score'].value_counts()

print("Home Score Frequency:")
print(home_score_freq)
print("Away Score Frequency:")
print(away_score_freq)
```

Home Score Frequency:

home_score

```
1    674
0    581
2    545
3    263
4    150
5     61
6     40
7     20
8       8
```

Name: count, dtype: int64

Away Score Frequency:

away_score

```
1    803
0    803
2    402
3    201
4     81
5     33
6     19
```

Name: count, dtype: int64

From the frequency of number of goals, we can see that playing as a home team has higher chance of scoring higher number of goals.

Part 2: Inferential Statistics

We can break our research questions down further to be more specific into these two questions:

- Does location of match correlates with home team winning?
- Does having a penalty kick in a match guarantees a win?

Correlation Analysis

```
In [115... from scipy.stats import chi2_contingency
from scipy.stats import pointbiseri...

no_duplicates_final = no_duplicates_final.copy()
# Derived variables
no_duplicates_final.loc[:, 'home_win'] = no_duplicates_final['home_score'] > no_du...
no_duplicates_final.loc[:, 'home_draw'] = no_duplicates_final['home_score'] == no_d...
no_duplicates_final.loc[:, 'home_advantage'] = (no_duplicates_final['neutral'] == F...

location_outcome = pd.crosstab(no_duplicates_final['neutral'], no_duplicates_final[...
print("Location vs. Home Team Win")
print(location_outcome)
print()
chi2_loc, p_loc, _, _ = chi2_contingency(location_outcome)
print(f"Chi-square: {chi2_loc}, p-value: {p_loc}\n")
```

```
# Ensure home_win is also a binary variable (1 for home win, 0 for not)
no_duplicates_final['home_win'] = no_duplicates_final['home_win'].astype(int)

# Calculate the point-biserial correlation between home advantage and home team win
correlation, p_value = pointbiserialr(no_duplicates_final['home_advantage'], no_duplicates_final['home_win'])

print(f"Correlation between home advantage and home team win: {correlation}")
```

Location vs. Home Team Win

home_win	False	True
neutral		
False	223	326
True	990	803

Chi-square: 35.27842944670624, p-value: 2.8577978437254603e-09

Correlation between home advantage and home team win: 0.12374151423432937

Answer

1. "Does location of match correlates with home team winning"

We did the cross-tabulation and chi-square test of independence.

Cross tabulation result: We found out that the frequency of home team winning when they're playing in non-neutral ground is higher, which means that there's a possibility that location correlates with winning the match.

Chi-Square result: We did a Chi-square test on the same variables as well to support our hypothesis, and it is true that they are correlated. With the p-value being **2.85e-09**, means that our p-value < 0.05 and we have to reject the null hypothesis. Which means that location is correlated to winning team.

Correlation The correlation between home advantage and home team winning is a weak positive value. **0.12374** shows that there may be a little bit correlation between home advantage and home team winning.

All this came to a result that playing as a home team with home advantage increases the chance of winning the match. Although it may not always be the case to wider external factors such as current team and player conditions, based on statistics, home team will have higher chance of winning.

In [116...

```
# For the 2nd research question, we will need to adjust our dataset
no_duplicates_final_with_pen = final_merged_data.copy()
# drop duplicates where penalty exists
no_duplicates_final_with_pen = no_duplicates_final_with_pen.drop_duplicates()
# Keep only matches where penalty happens
no_duplicates_final_with_pen['team_winner'] = np.where(
    no_duplicates_final_with_pen['home_score'] > no_duplicates_final_with_pen['away_score'],
    no_duplicates_final_with_pen['home_team'], # Assign home_team value
    no_duplicates_final_with_pen['away_team'] # Assign Non
```

```

)

# is_winner will need to be an integer value so we can use it as a correlation variable
no_duplicates_final_with_pen['is_winner'] = np.where(
    no_duplicates_final_with_pen['team_winner'] == no_duplicates_final_with_pen['team_winner'],
    1, # Assign 1 if they match
    0 # Assign 0 if they do not match
)

# change the penalty value to 1 and 0
no_duplicates_final_with_pen['penalty'] = no_duplicates_final_with_pen['penalty'].astype(int)

no_duplicates_final_with_pen['is_winner'] = pd.to_numeric(no_duplicates_final_with_pen['is_winner'], errors='coerce')
no_duplicates_final_with_pen['penalty'] = pd.to_numeric(no_duplicates_final_with_pen['penalty'], errors='coerce')

# Derived variables
no_duplicates_final_with_pen.loc[:, 'home_win'] = no_duplicates_final_with_pen['home_team_winner']
no_duplicates_final_with_pen.loc[:, 'home_draw'] = no_duplicates_final_with_pen['home_team_draw']

penalty_outcome = pd.crosstab(no_duplicates_final_with_pen['penalty'], no_duplicates_final_with_pen['is_winner'])
print("Penalty vs. Match Outcome")
print(penalty_outcome)
print()
chi2_pen, p_pen, _, _ = chi2_contingency(penalty_outcome)
print(f"Chi-square: {chi2_pen}, p-value: {p_pen}")

penalty_correlation = no_duplicates_final_with_pen['penalty'].corr(no_duplicates_final_with_pen['is_winner'])
print("correlation value between having a penalty kick and winning the match:", penalty_correlation)

```

Penalty vs. Match Outcome

is_winner	0	1
penalty		
0	584	1721
1	140	293

Chi-square: 8.817342022131562, p-value: 0.0029838098562191607

correlation value between having a penalty kick and winning the match: -0.057883018932532955

2. Does having a penalty guarantees a win? We may have found an opposite result based on these analysis. The cross-tabulation and chi-square test shows that having penalty kicks and winning the match may have correlation, but the pearson correlation test shows that there's weak close to no correlation.

Now the question is, WHY??

As we know, Pearson correlation measures the linear relationship between two continuous variables. When the variables are binary, it measures how well the two variables change together in a linear fashion. The correlation value of **-0.057883** shows that there's weak inverse linear relationship, but it doesn't mean that there is no relationship at all. In fact, this suggests that the pattern is not strong enough to show a meaningful association using Pearson's method.

Through the cross-tabulation, we can see clearly from the frequency of having a penalty and match outcomes shows that having a penalty increases the chance of winning. Again, this may not guarantee a win, but having the ratio of 140:293 shows penalty plays a role in a team winning, resulting in more than **66%** winrate.

In addition, the Chi-square test of independence resulted in a p-value = **0.0029838**, which is less than the common threshold of 0.05. This means that there is a statistically significant association between the two variables, penalty and winning the match.

Key takeaway

While the correlation is weak and negative, the Chi-square test and cross-tabulation both suggest that penalties do have an effect on winning, but the relationship is more complex and may not be captured well by a simple linear correlation.

Group Comparisons

T-Test: Comparing Two Groups

We will start with the first question:

Does location of match correlates with home team winning?

```
In [117... from scipy.stats import ttest_ind

# T-test: Compare average home_score vs away_score
ttest_home_away = ttest_ind(
    no_duplicates_final['home_score'],
    no_duplicates_final['away_score']
)

print("T-Test: Home Score vs Away Score")
print(f"T-statistic: {ttest_home_away.statistic}, p-value: {ttest_home_away.pvalue}")

# T-test: Compare average goals in neutral vs non-neutral matches
neutral_goals = no_duplicates_final.loc[no_duplicates_final['neutral'], 'home_score']
non_neutral_goals = no_duplicates_final.loc[~no_duplicates_final['neutral'], 'home_score']

ttest_neutral = ttest_ind(neutral_goals, non_neutral_goals)

print("T-Test: Goals in Neutral vs Non-Neutral Matches")
print(f"T-statistic: {ttest_neutral.statistic}, p-value: {ttest_neutral.pvalue}\n")
```

T-Test: Home Score vs Away Score

T-statistic: 12.776211254801284, p-value: 7.450669918655214e-37

T-Test: Goals in Neutral vs Non-Neutral Matches

T-statistic: 0.48205307502551825, p-value: 0.6298068015530757

Interpretation of the T-Test Results for Home vs Away Matches

T-Statistic: A large T-statistic indicates that the means of the two groups (home_score and away_score) are significantly different.

P-Value: The extremely small p-value ($p < 0.05$) provides strong evidence to reject the null hypothesis, which assumes no difference in the average scores between home and away teams.

Key Takeaway:

This result suggests that the average home score is significantly higher than the average away score, indicating a statistical relationship between the location (home vs. away) and the likelihood of the home team scoring more goals.

Interpretation of the T-Test Results for Neutral vs Non-Neutral Matches:

T-Statistic: A low T-statistic indicates that the means of the two groups (total goals in neutral vs. non-neutral matches) are not significantly different.

P-Value: The relatively large p-value ($p=0.630$) provides insufficient evidence to reject the null hypothesis, which assumes no difference in the average number of goals between neutral and non-neutral matches.

Key Takeaway:

This result suggests that there is no statistically significant difference in the average number of goals scored in matches played at neutral locations versus non-neutral locations. While neutral venues may impact the likelihood of a team winning, they do not appear to significantly affect the total number of goals scored in a match.

We will now move onto the second question:

Does having a penalty kick in a match guarantees a win?

In [118...

```
# Create a copy of the dataframe
penalty_data = no_duplicates_final_with_pen.copy()

# Create a binary column for match win (1 = win, 0 = not win)
penalty_data['match_won'] = (penalty_data['home_score'] > penalty_data['away_score'])

# Create two groups based on penalty kicks
with_penalty = penalty_data[penalty_data['penalty'] == 1]['match_won']
without_penalty = penalty_data[penalty_data['penalty'] == 0]['match_won']

# Perform the T-test
ttest_penalty = ttest_ind(with_penalty, without_penalty)

# Print the results
```

```
print("T-Test: Penalty Kick vs Match Win")  
print(f"T-statistic: {ttest_penalty.statistic}, p-value: {ttest_penalty.pvalue}\n")
```

T-Test: Penalty Kick vs Match Win

T-statistic: 0.9299795976986696, p-value: 0.35246369158104995

Interpretation of the T-Test Results:

T-Statistic: A small T-statistic (0.93) indicates that the means of the two groups (win rates for matches with and without penalties) are not significantly different.

P-Value: The relatively large p-value ($p=0.352$) provides insufficient evidence to reject the null hypothesis, which assumes no difference in the average win rates between matches with and without penalties.

Key Takeaway

This result suggests that having a penalty kick in a match does not guarantee a win. The average win rates for matches with and without penalties are not significantly different, indicating that penalties alone do not have a statistically significant impact on match outcomes. External factors such as team strategy, timing of the penalty, or match conditions may play a larger role in determining the outcome.

Regression Analysis

To determine the best regression model for the first question, we need to better understand our variables.

Does the location of a match correlate with the home team winning?

Variables:

- Dependent Variable: home_win (binary: 1 for a home win, 0 otherwise).
- Independent Variables: neutral (binary: neutral location or not), potentially others like home_score and away_score.

Model Suitability:

- Simple Linear Regression: Predicts a continuous dependent variable from one independent variable. Not suitable here since home_win is binary.
- Multiple Linear Regression: Predicts a continuous dependent variable from multiple independent variables. Also unsuitable due to the binary nature of home_win.
- Logistic Regression: Predicts a binary dependent variable based on one or more independent variables. Ideal for this question.

Selected Model: Logistic Regression

Logistic regression is best suited for analyzing the relationship between the location (neutral or not) and the probability of the home team winning.

```
In [119... import statsmodels.api as sm

logit_data = no_duplicates_final.copy()

# Create dependent variable: Did the designated home team win?
logit_data['home_win'] = (logit_data['home_score'] > logit_data['away_score']).astype(int)

# Create independent variable: Neutral venue (binary)
logit_data['neutral'] = logit_data['neutral'].astype(int) # Ensure it's numeric

# Define independent and dependent variables
X = logit_data[['neutral']] # Independent variable: Neutral venue
X = sm.add_constant(X) # Add constant term for intercept
y = logit_data['home_win'] # Dependent variable: Home team win

# Fit Logistic regression model
logit_model = sm.Logit(y, X).fit()

# Print results summary
print(logit_model.summary())
```

Optimization terminated successfully.

Current function value: 0.684941

Iterations 4

Logit Regression Results						
Dep. Variable:	home_win	No. Observations:	2738			
Model:	Logit	Df Residuals:	2736			
Method:	MLE	Df Model:	1			
Date:	Sat, 14 Dec 2024	Pseudo R-squ.:	0.01140			
Time:	20:11:26	Log-Likelihood:	-1875.4			
converged:	True	LL-Null:	-1897.0			
Covariance Type:	nonrobust	LLR p-value:	4.820e-11			
	coef	std err	z	P> z	[0.025	0.975]
const	0.4134	0.081	5.091	0.000	0.254	0.573
neutral	-0.6011	0.092	-6.516	0.000	-0.782	-0.420

The logistic regression analysis explores the impact of neutral venues on the likelihood of the home team (designated) winning. The coefficient for neutral is -0.6011, indicating that matches played at neutral venues reduce the odds of the home team winning. This effect is statistically significant, with a p-value of less than 0.001, meaning neutral venues influence match outcomes significantly.

The odds ratio for neutral matches ($e^{-0.6011} \approx 0.548$) suggests that the odds of the designated home team winning at a neutral venue are approximately 54.8% of the odds in non-neutral matches. The intercept (0.4134) reflects the baseline odds of the home team winning when the match is not played at a neutral venue.

While the Pseudo R-squared value is low (0.0114), indicating other factors contribute to match outcomes, the results confirm that neutral venues reduce the likelihood of the home team winning, highlighting the importance of home advantage.

We will now work on the second question.

Does having a penalty kick in a match guarantee a win?

Variables:

- Dependent Variable: home_win (binary: 1 for a home win, 0 otherwise).
- Independent Variable: penalty (binary: 1 if a penalty occurs, 0 otherwise).

Model Suitability:

- Simple Linear Regression: Predicts a continuous dependent variable from one independent variable. Not suitable here since home_win is binary.
- Multiple Linear Regression: Predicts a continuous dependent variable from multiple independent variables. Also unsuitable due to the binary nature of home_win.
- Logistic Regression: Predicts a binary dependent variable based on one or more independent variables. Ideal for this question as both the dependent and independent variables are binary.

Selected Model: Logistic Regression

Logistic regression is best suited for analyzing the relationship between penalties and the probability of the home team winning. It allows us to quantify how penalties influence the likelihood of winning, providing insights through coefficients, p-values, and odds ratios.

```
In [120... penalty_data = no_duplicates_final.copy()

# Create binary dependent variable: Did the home team win?
penalty_data['home_win'] = (penalty_data['home_score'] > penalty_data['away_score'])

# Ensure the penalty column is numeric
penalty_data['penalty'] = penalty_data['penalty'].astype(int)

# Define independent and dependent variables
X = penalty_data[['penalty']] # Independent variable: Penalty
X = sm.add_constant(X) # Add constant term for intercept
y = penalty_data['home_win'] # Dependent variable: Home team win

# Fit logistic regression model
penalty_model = sm.Logit(y, X).fit()

# Print results summary
print(penalty_model.summary())
```

Optimization terminated successfully.
 Current function value: 0.692681
 Iterations 3

Logit Regression Results

Dep. Variable:	home_win	No. Observations:	2738
Model:	Logit	Df Residuals:	2736
Method:	MLE	Df Model:	1
Date:	Sat, 14 Dec 2024	Pseudo R-squ.:	0.0002280
Time:	20:11:26	Log-Likelihood:	-1896.6
converged:	True	LL-Null:	-1897.0
Covariance Type:	nonrobust	LLR p-value:	0.3524

	coef	std err	z	P> z	[0.025	0.975]
const	-0.0651	0.042	-1.562	0.118	-0.147	0.017
penalty	0.0974	0.105	0.930	0.352	-0.108	0.303

The logistic regression analysis investigates whether having a penalty kick impacts the likelihood of the home team winning. The coefficient for penalty is 0.0974, suggesting a minor increase in the odds of the home team winning when a penalty occurs. However, the effect is not statistically significant ($p = 0.352$), as the p-value exceeds the standard threshold ($\alpha = 0.05$).

The odds ratio ($e^{0.0974} \approx 1.10$) indicates a very small increase in the odds of winning with a penalty. The intercept (-0.0651) reflects the baseline odds of the home team winning when no penalty occurs. Additionally, the model's Pseudo R-squared value (0.000228) shows that penalties do not explain the variance in match outcomes.

Overall, the results suggest that having a penalty kick does not significantly affect the likelihood of winning, highlighting the importance of other factors in determining match outcomes.

Part 4: Documentation

Our analysis aimed to answer the main research question which is to explore trends in international football over time, focusing on home advantage, goal-scoring patterns, and the evolving strength of teams.

We investigated two specific questions to address these themes:

1. Does the location of a match correlate with the home team winning?
2. Does having a penalty kick in a match guarantee a win?

Descriptive Analysis

The mean home_score (1.67) was higher than the mean away_score (1.20), suggesting a potential home advantage. The median score for both home_score and away_score was 1,

with the most common home_score being 1 and the most common away_score being 0. This highlights the defensive nature of many matches. The avg goal timing (minute mean = 28.03, median = 21) showed a balanced spread of goals across the match.

Categorical Analysis

The most common home_score was 1, followed by 0, while away_score was most often 1 or 0. This shows that international matches have a more defensive match style. Cross-tabulations also showed home teams won more in non-neutral venues (326 vs. 223) and a 66% win rate with penalties. Additionally penalties do not guarantee victory.

Inferential Statistics

- **Does the location of a match correlate with the home team winning?**

A weak positive correlation ($r = 0.1237$) was found between home_advantage (non-neutral venues) and home_win, indicating a slight increase in the likelihood of winning when playing at home.

Logistic regression confirmed that the location of a match influences its outcome significantly. Matches at neutral venues reduce the likelihood of the home team winning, with an odds ratio of approximately 0.548. This finding underscores the importance of home advantage, some reasons include external factors such as crowd support and familiarity with the venue.

A chi-square test of independence ($p < 0.001$) further validated this relationship. The weak correlation aligns with these results, suggesting that while location matters, other factors also influence match outcomes.

- **Does having a penalty kick in a match guarantee a win?**

For penalties, the correlation between penalties and match outcomes was significantly negative ($r = -0.0579$), this shows that penalties do not guarantee a win.

Logistic regression and chi-square tests revealed a quite a complex relationship. Although penalties were associated with a statistically significant relationship to match outcomes ($p = 0.0029$), the correlation ($r = -0.0579$) showed a weak inverse linear relationship. This means that penalties play a role in specific moments but do not guarantee a win.

Some key takeaways from this as a summary:

- **Home Advantage:** Location is a significant factor in international football, with home teams having a measurable advantage in non-neutral venues.
- **Goal-Scoring Patterns:** Penalties are impactful but not a guarantee of victory, as their influence depends on other factors, such as how well a team is playing at that moment or when the penalty was given or scored.

- Evolving Trends: While location remains a critical factor, the role of penalties and other match-specific factors is not very predictable.

Limitations and Future Directions The low R-squared values suggests that unmeasured factors, such as team strength, tournament context, or player-level data, can also influence the outcome of a game. This should be measured in additional future research.

Data Visualizations

Part 1: Essential Plot Types

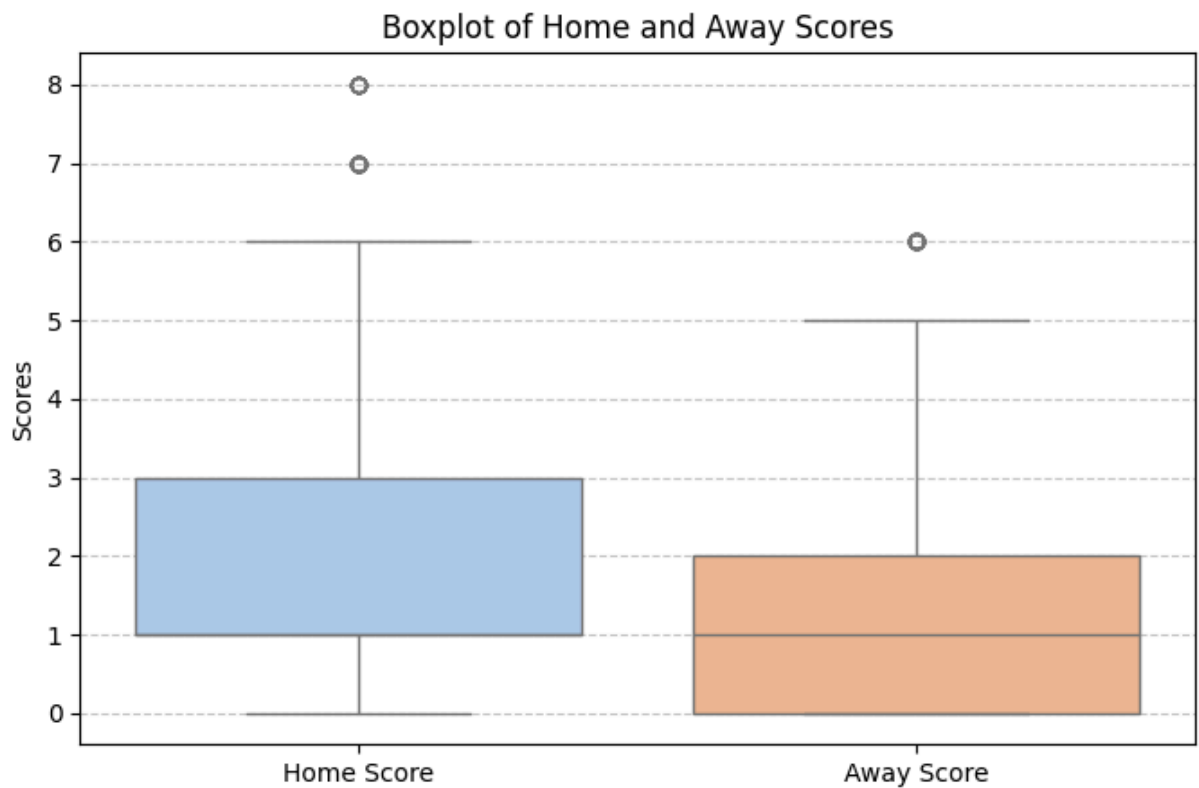
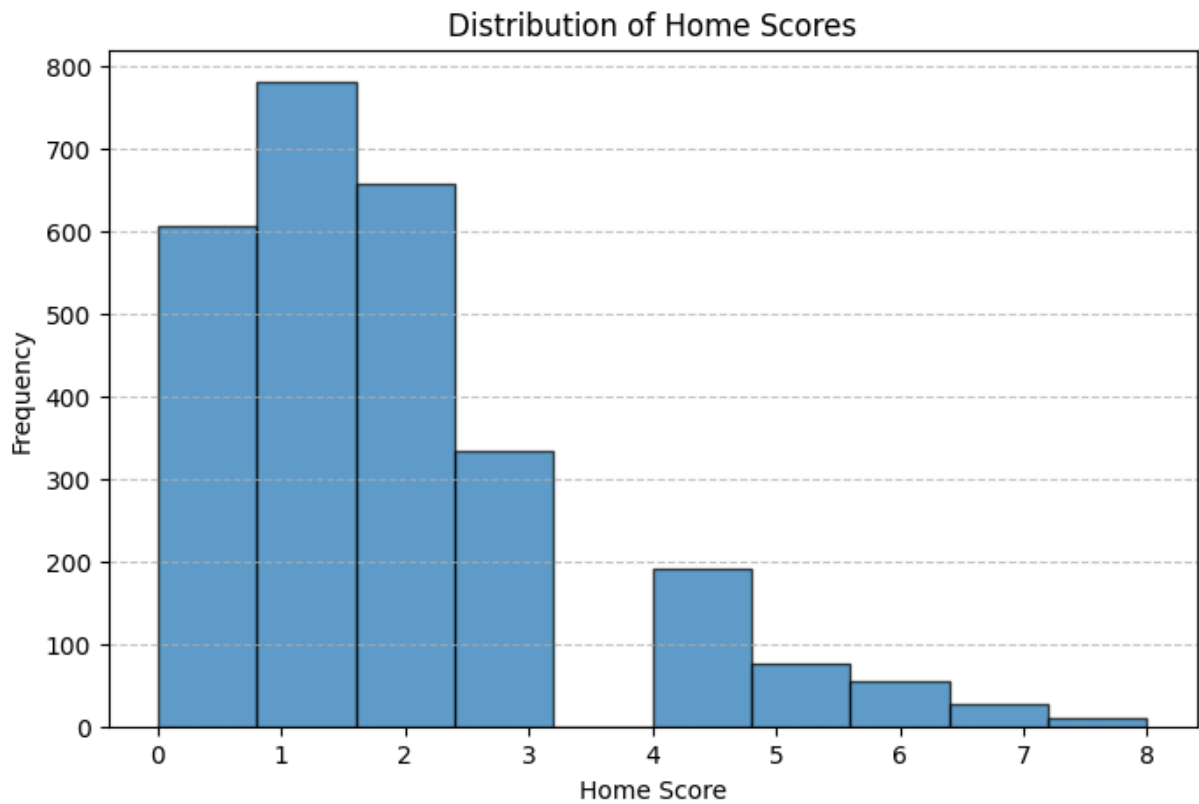
1. Distribution Analysis

```
In [121... import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [122... # Visualizations to understand the frequency distribution of home_scores and how ho

# Histogram: Distribution of `home_score`
plt.figure(figsize=(8, 5))
plt.hist(no_duplicates_final['home_score'], bins=10, edgecolor='black', alpha=0.7)
plt.title('Distribution of Home Scores')
plt.xlabel('Home Score')
plt.ylabel('Frequency')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

# Boxplot: Comparison of `home_score` and `away_score`
plt.figure(figsize=(8, 5))
sns.boxplot(data=no_duplicates_final[['home_score', 'away_score']], palette="pastel")
plt.title('Boxplot of Home and Away Scores')
plt.ylabel('Scores')
plt.xticks([0, 1], ['Home Score', 'Away Score'])
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Histogram:

- Displays the frequency distribution of home_score.
- Customizations include bins, gridlines, and alpha transparency for aesthetics.

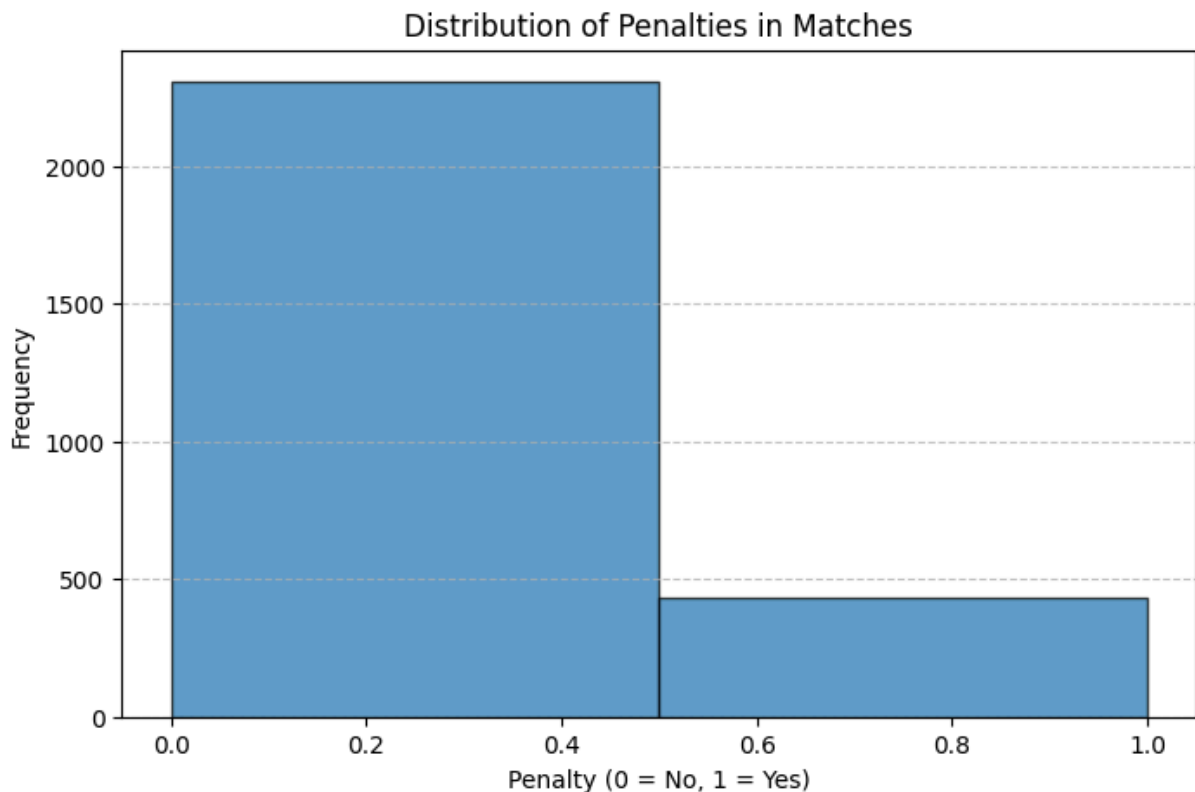
Boxplot:

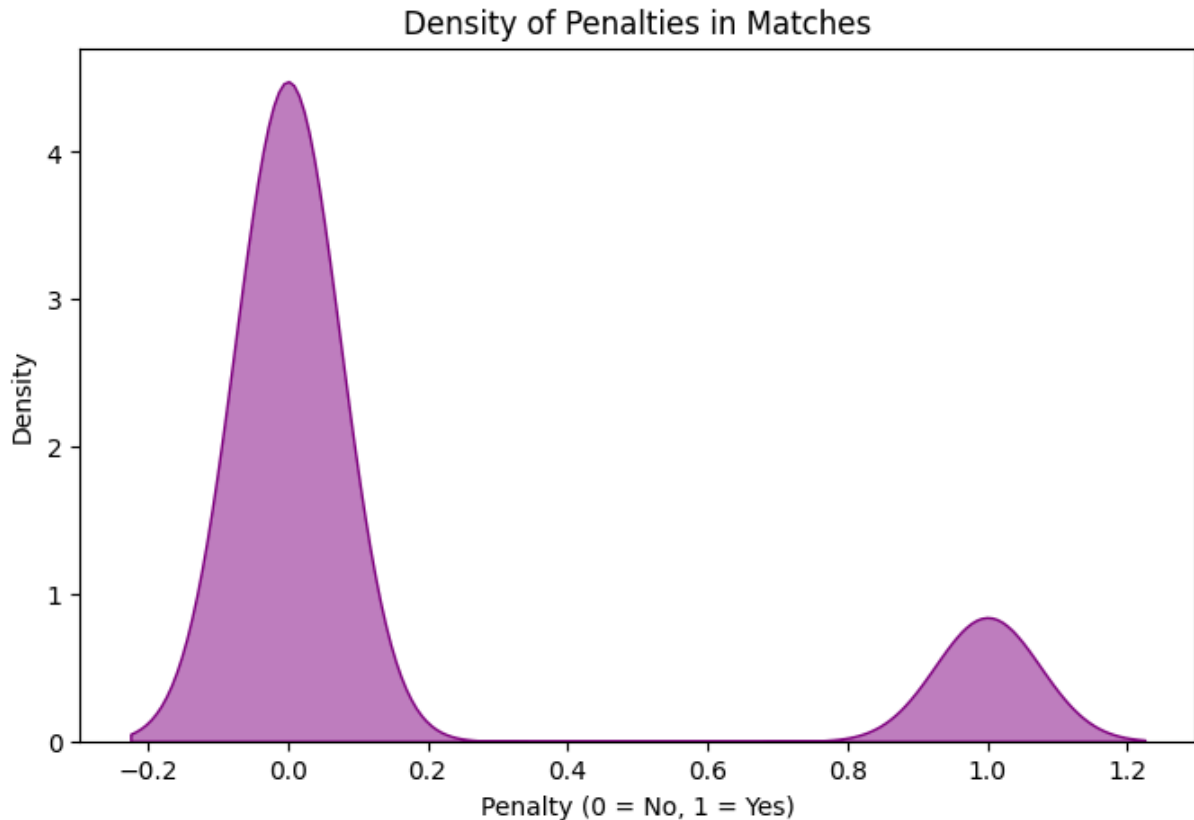
- Compares home_score and away_score distributions.
- Useful for identifying outliers and assessing score variability.

```
In [123... # Visualizations to understand the effects of penalties in a game

# Histogram: Frequency of penalties
plt.figure(figsize=(8, 5))
plt.hist(no_duplicates_final_with_pen['penalty'], bins=2, edgecolor='black', alpha=
plt.title('Distribution of Penalties in Matches')
plt.xlabel('Penalty (0 = No, 1 = Yes)')
plt.ylabel('Frequency')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

# KDE Plot: Density of penalties in matches
plt.figure(figsize=(8, 5))
sns.kdeplot(data=no_duplicates_final_with_pen, x='penalty', fill=True, color='purple
plt.title('Density of Penalties in Matches')
plt.xlabel('Penalty (0 = No, 1 = Yes)')
plt.ylabel('Density')
plt.show()
```





Histogram:

- Displays the frequency of penalties (penalty = 0 or 1) across all matches.
- Useful to see how often penalties occur in the dataset.

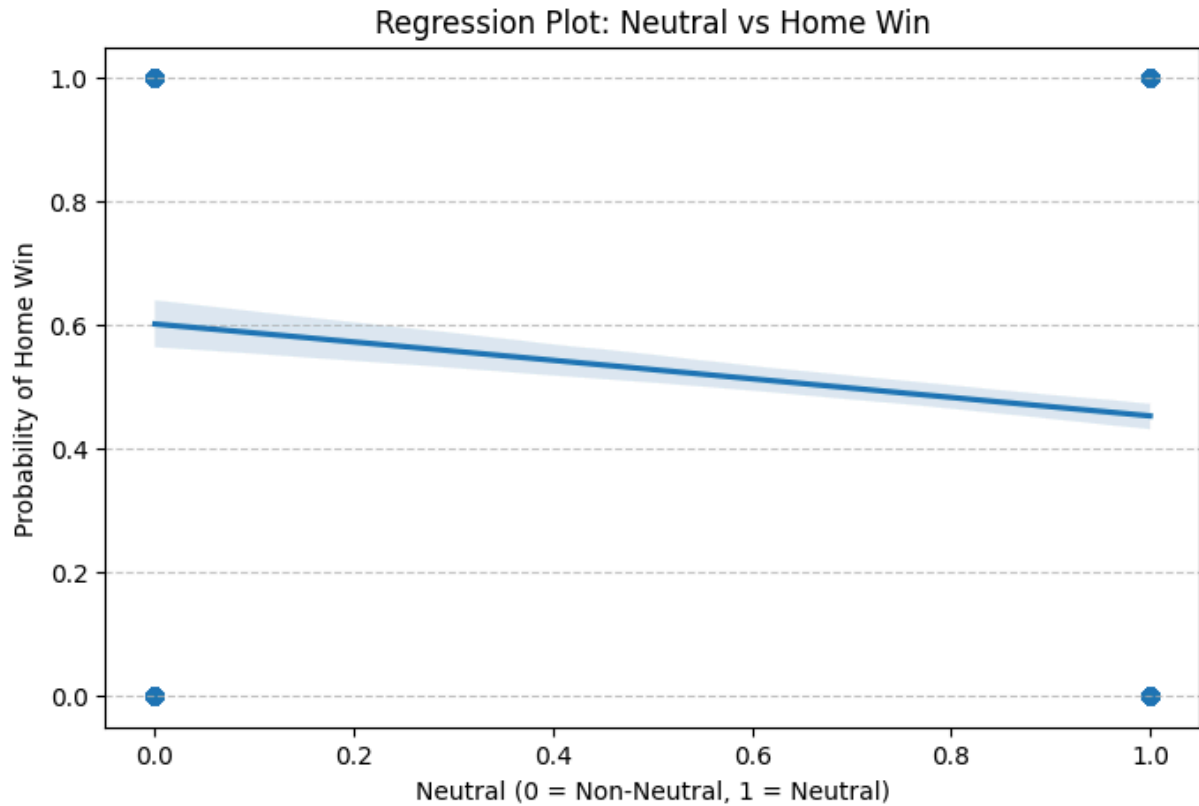
KDE Plot:

- Provides a smooth density curve for penalties (penalty).
- Helps to visualize the probability distribution of penalties in matches.

2. Relationship Analysis

```
In [124... # Visualizations to analyze the relationship between the location of a match and th

# Regression Plot: Neutral vs Home Win
plt.figure(figsize=(8, 5))
sns.regplot(x='neutral', y='home_win', data=no_duplicates_final, logistic=True, sca
plt.title('Regression Plot: Neutral vs Home Win')
plt.xlabel('Neutral (0 = Non-Neutral, 1 = Neutral)')
plt.ylabel('Probability of Home Win')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Regression Plot:

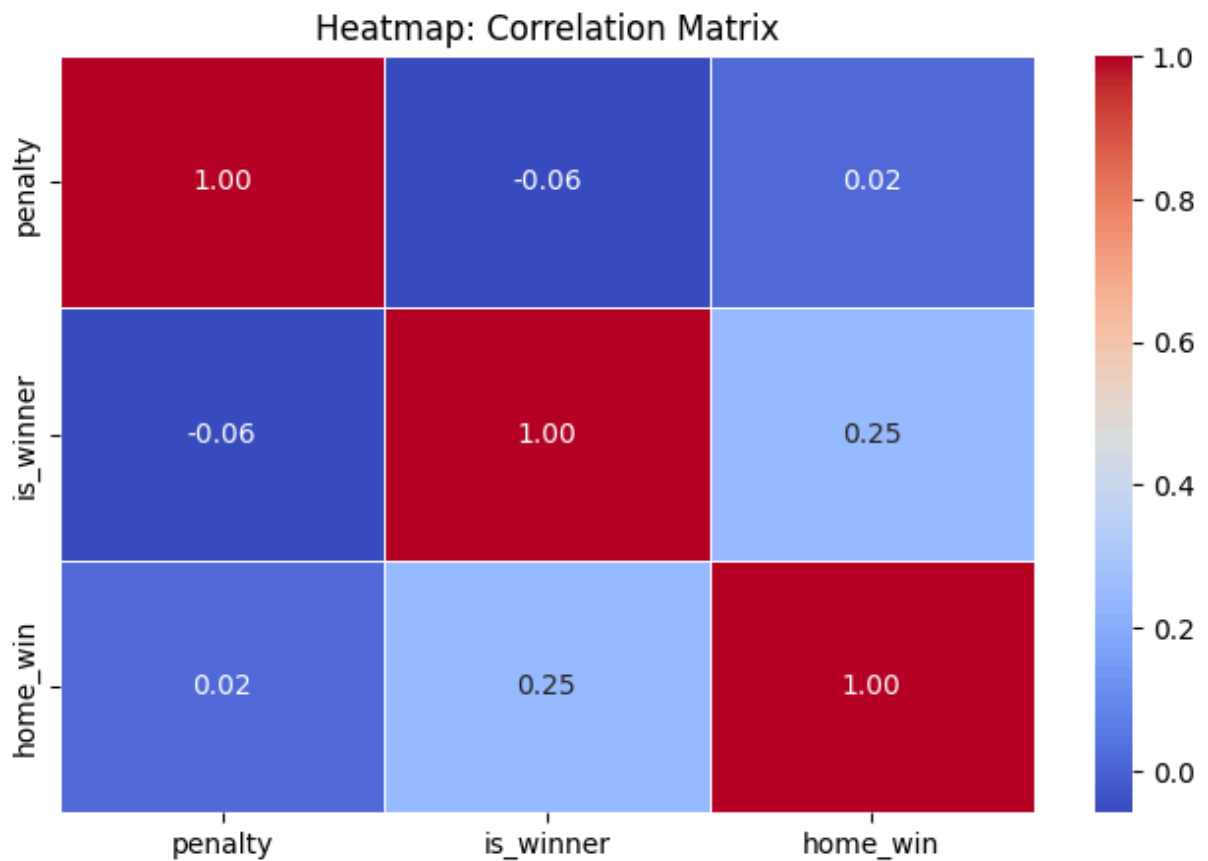
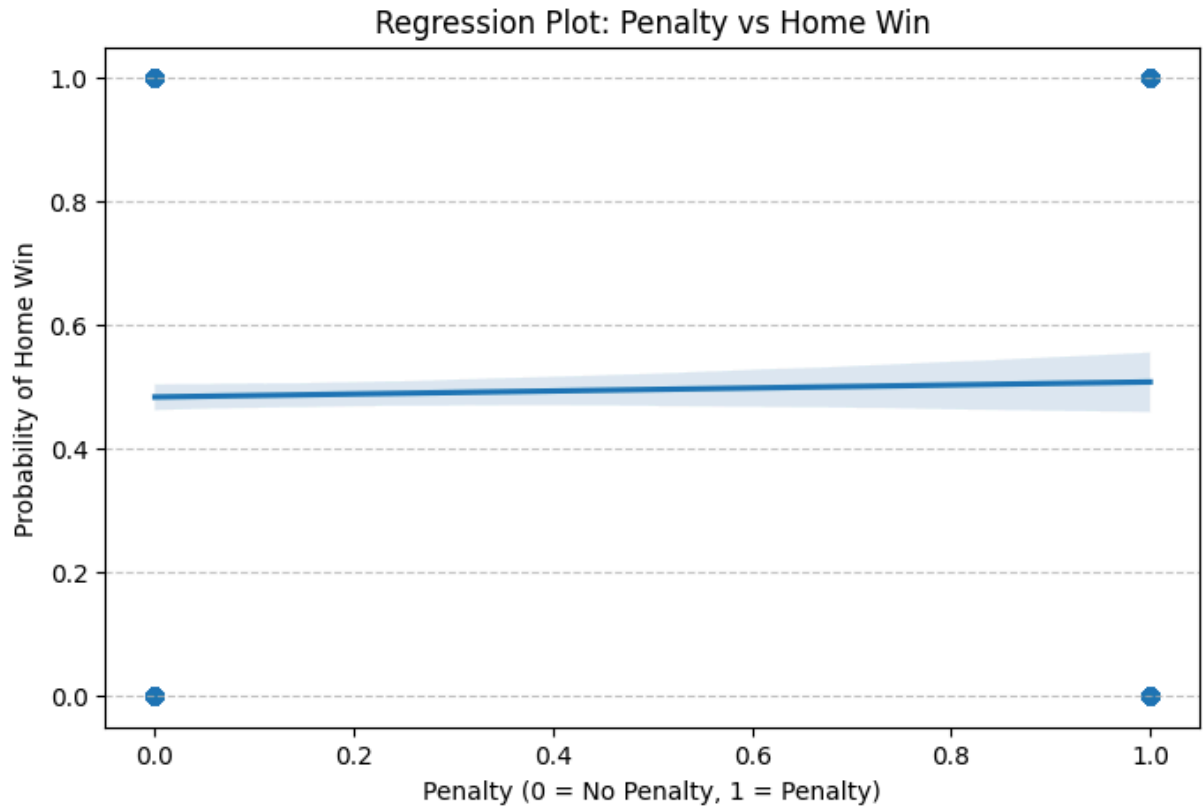
- A scatter plot with a fitted regression line (logistic regression in this case) to show the trend in the probability of a home win based on the venue type.

In [125...

```
# Visualizations to understand the relationship between penalties and the likelihood of a home win

# Regression Plot: Penalty vs Home Win
plt.figure(figsize=(8, 5))
sns.regplot(x='penalty', y='home_win', data=no_duplicates_final_with_pen, logistic=True)
plt.title('Regression Plot: Penalty vs Home Win')
plt.xlabel('Penalty (0 = No Penalty, 1 = Penalty)')
plt.ylabel('Probability of Home Win')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

# Heatmap: Correlation Matrix
plt.figure(figsize=(8, 5))
corr_matrix = no_duplicates_final_with_pen[['penalty', 'is_winner', 'home_win']].corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Heatmap: Correlation Matrix')
plt.show()
```

Regression Plot:

- A scatter plot with a logistic regression line showing the probability of a home win based on whether a penalty occurred.

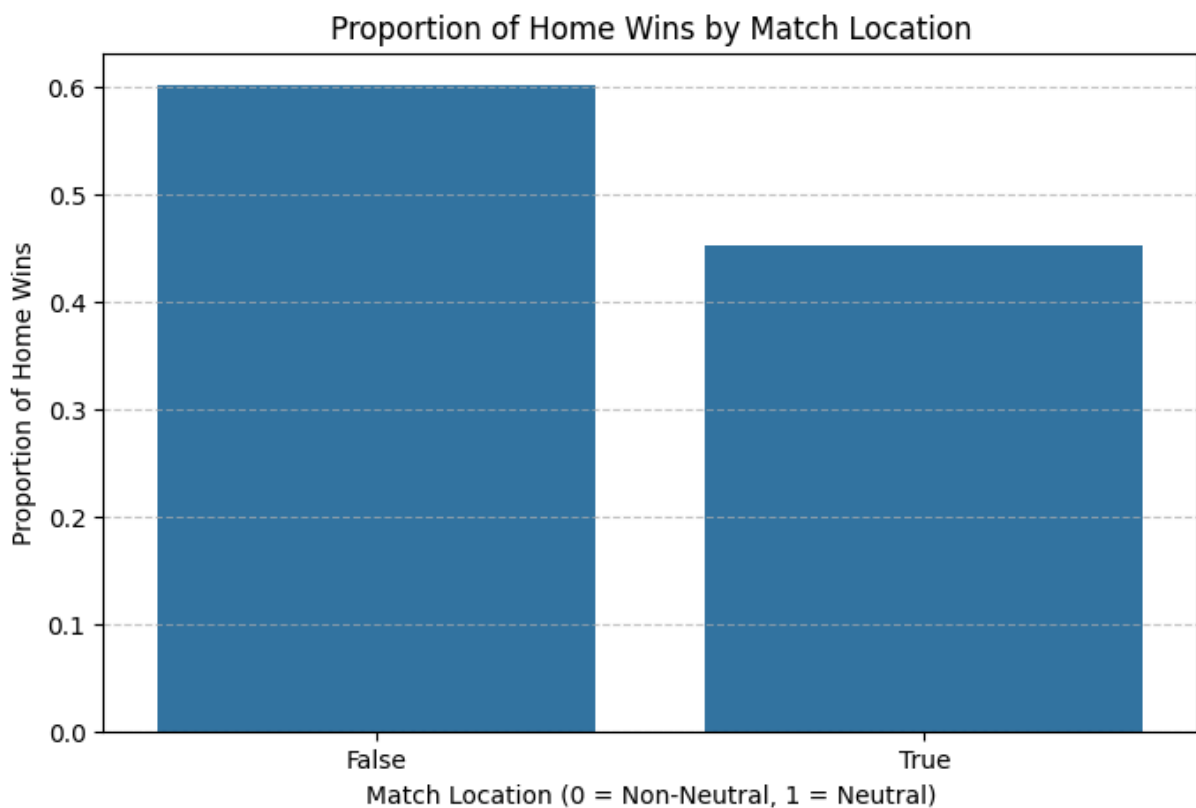
Heatmap:

- Highlights the correlation strengths between variables (penalty, is_winner, and home_win), making it easy to see which variables are related and to what extent.

Comparison Analysis

```
In [126... # Visualization - comparison analysis to analyze match outcomes depending on match

# Grouped Bar Plot: Wins and Losses by Location
plt.figure(figsize=(8, 5))
sns.barplot(x='neutral', y='home_win', data=no_duplicates_final, errorbar=None)
plt.title('Proportion of Home Wins by Match Location')
plt.xlabel('Match Location (0 = Non-Neutral, 1 = Neutral)')
plt.ylabel('Proportion of Home Wins')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



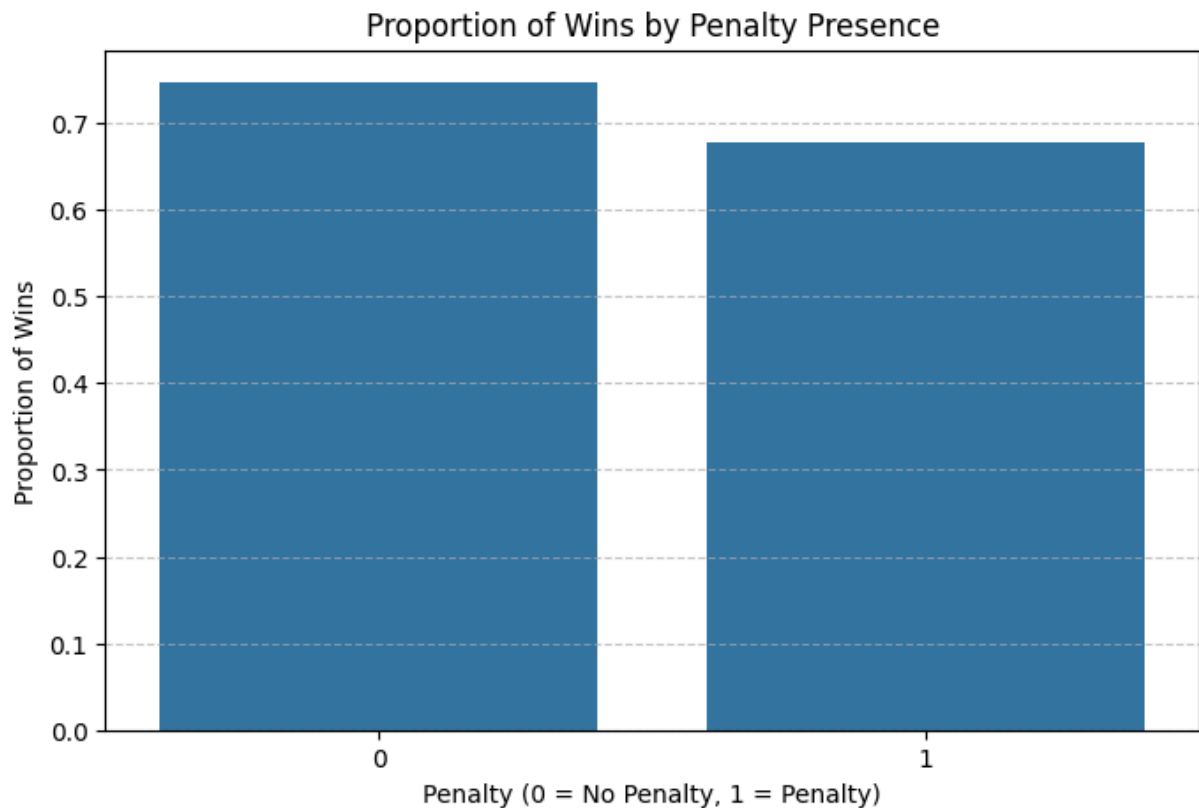
Grouped Bar Plot:

- Compares the proportion of home wins for neutral and non-neutral matches. Useful for showing trends in winning probabilities by match location.

```
In [127... # Comparison analysis to analyze effect of penalties on the outcome of a match

# Grouped Bar Plot: Proportion of Wins by Penalty Presence
plt.figure(figsize=(8, 5))
```

```
sns.barplot(x='penalty', y='is_winner', data=no_duplicates_final_with_pen, errorbar
plt.title('Proportion of Wins by Penalty Presence')
plt.xlabel('Penalty (0 = No Penalty, 1 = Penalty)')
plt.ylabel('Proportion of Wins')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Grouped Bar Plot:

- Highlights the proportion of matches won for both penalty and no-penalty scenarios.
- Useful for assessing the winning likelihood based on penalty occurrence.

Composition Plots

In [128...

```
# Variables: Neutral and Home Win Status
neutral_matches = no_duplicates_final[(no_duplicates_final['neutral'] == 1)]
non_neutral_matches = no_duplicates_final[(no_duplicates_final['neutral'] == 0)]

# Proportions for neutral matches
neutral_win_counts = neutral_matches['home_win'].value_counts()
neutral_win_labels = ['Win', 'Loss']

# Proportions for non-neutral matches
non_neutral_win_counts = non_neutral_matches['home_win'].value_counts()
non_neutral_win_labels = ['Win', 'Loss']

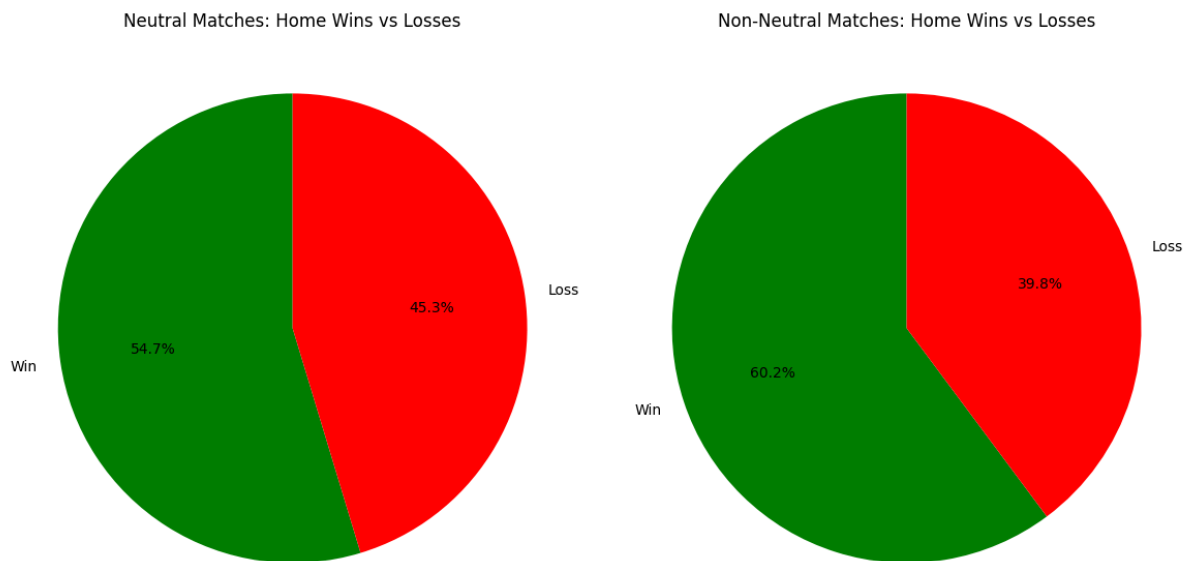
# Create Pie Charts
fig, axes = plt.subplots(1, 2, figsize=(12, 6))
axes[0].pie(neutral_win_counts, labels=neutral_win_labels, autopct='%1.1f%%', color
```

```

axes[0].set_title('Neutral Matches: Home Wins vs Losses')
axes[1].pie(non_neutral_win_counts, labels=non_neutral_win_labels, autopct='%1.1f%%')
axes[1].set_title('Non-Neutral Matches: Home Wins vs Losses')

plt.tight_layout()
plt.show()

```



Pie Chart:

- Two pie charts show the proportion of wins and losses for non-neutral and neutral matches.
- Highlights the dominance of wins in non-neutral matches compared to neutral matches.

In [129...

```

# Composition plot to analyze the effect of penalties on the outcome of a match

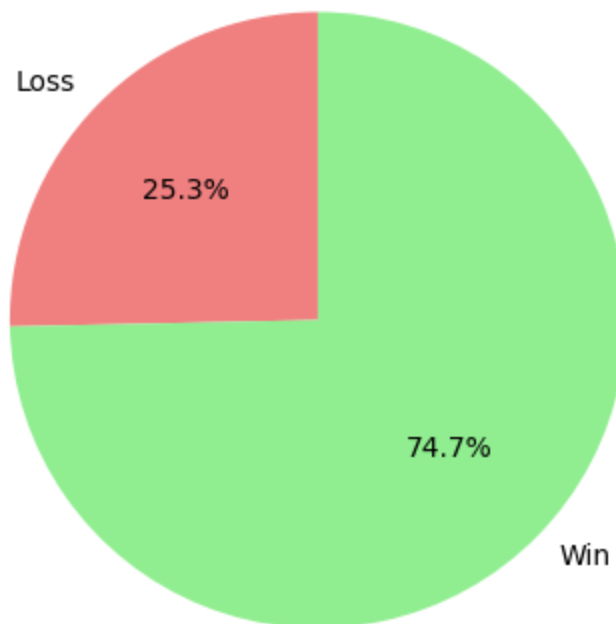
# Data Preparation
penalty_outcome = no_duplicates_final_with_pen.groupby('penalty')['is_winner'].value_counts()

# Pie Chart: Proportion of Wins and Losses for Matches Without Penalties
plt.figure(figsize=(8, 5))
no_penalty = penalty_outcome.loc[0] # No penalty matches
plt.pie(no_penalty, labels=['Loss', 'Win'], autopct='%1.1f%%', startangle=90, color=['red', 'green'])
plt.title('Match Outcomes for Matches Without Penalties')
plt.show()

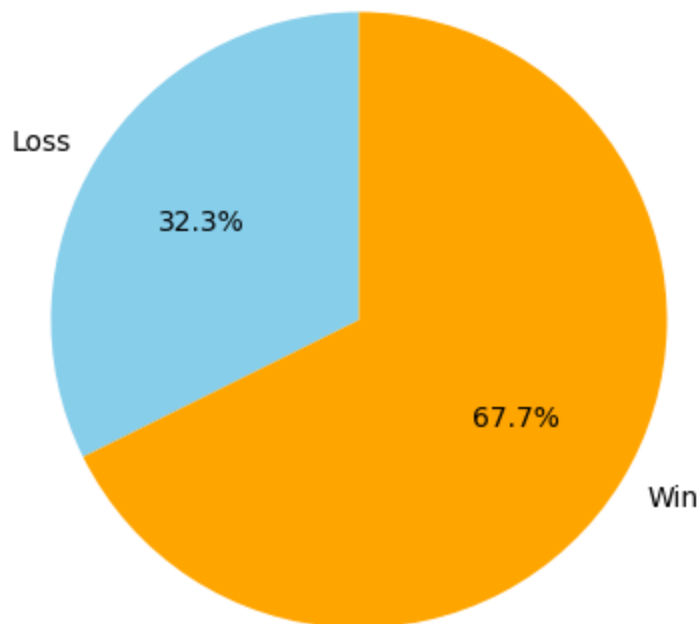
# Pie Chart: Proportion of Wins and Losses for Matches With Penalties
plt.figure(figsize=(8, 5))
with_penalty = penalty_outcome.loc[1] # Matches with penalties
plt.pie(with_penalty, labels=['Loss', 'Win'], autopct='%1.1f%%', startangle=90, color=['red', 'green'])
plt.title('Match Outcomes for Matches With Penalties')
plt.show()

```

Match Outcomes for Matches Without Penalties



Match Outcomes for Matches With Penalties



Pie Chart:

- Two separate pie charts show the proportion of wins and losses for matches without penalties and with penalties.

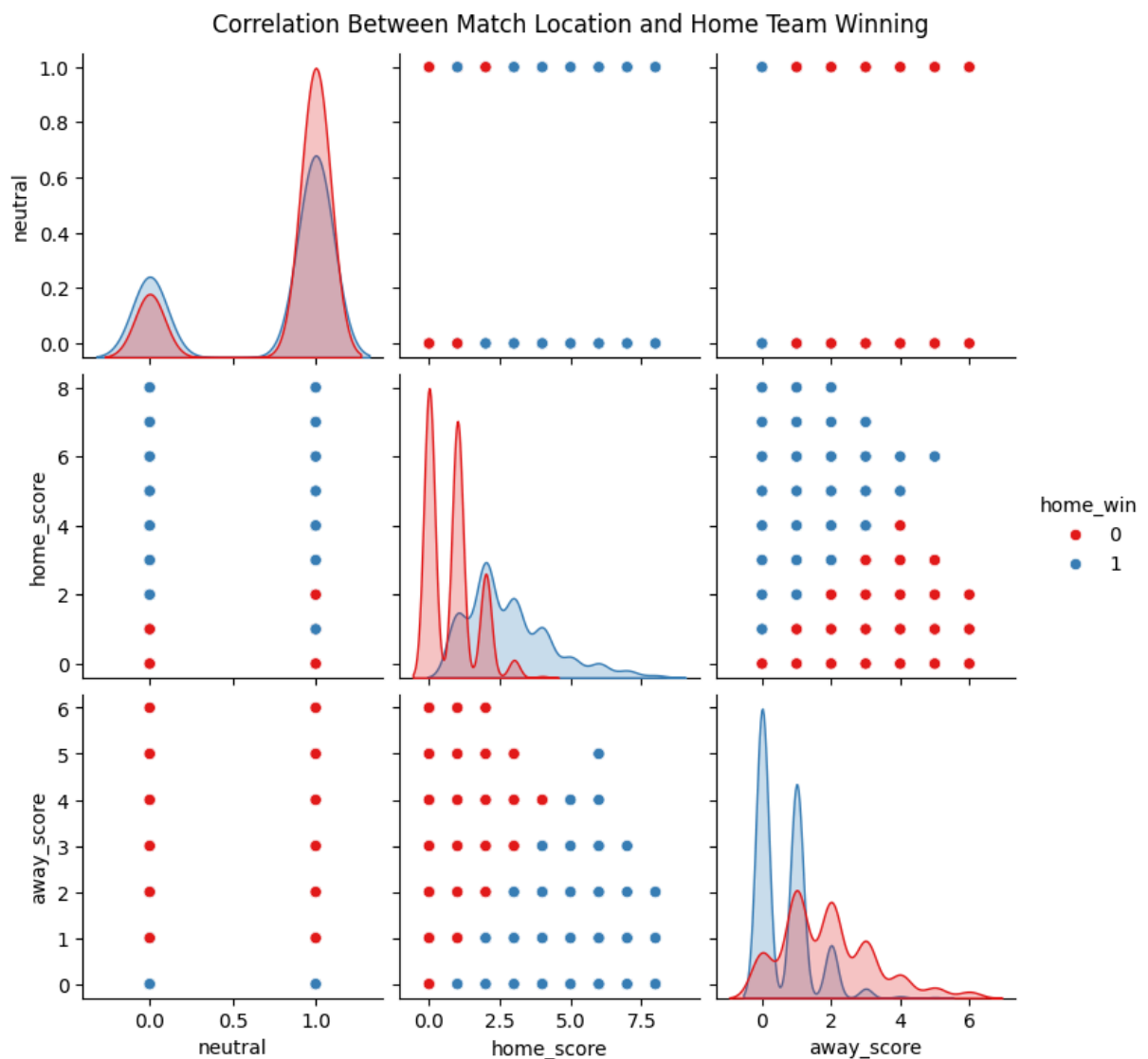
- Highlights how penalties influence match outcomes.

Multi-Variable Analysis

```
In [130... no_duplicates_final['neutral'] = no_duplicates_final['neutral'].astype(int) # Conv
no_duplicates_final['home_win'] = no_duplicates_final['home_win'].astype(int) # Co

# Select relevant columns
columns_to_plot = ['neutral', 'country', 'home_win', 'home_score', 'away_score']

# Pairplot
sns.pairplot(no_duplicates_final[columns_to_plot], hue='home_win', diag_kind='kde',
plt.suptitle('Correlation Between Match Location and Home Team Winning', y=1.02)
plt.show())
```



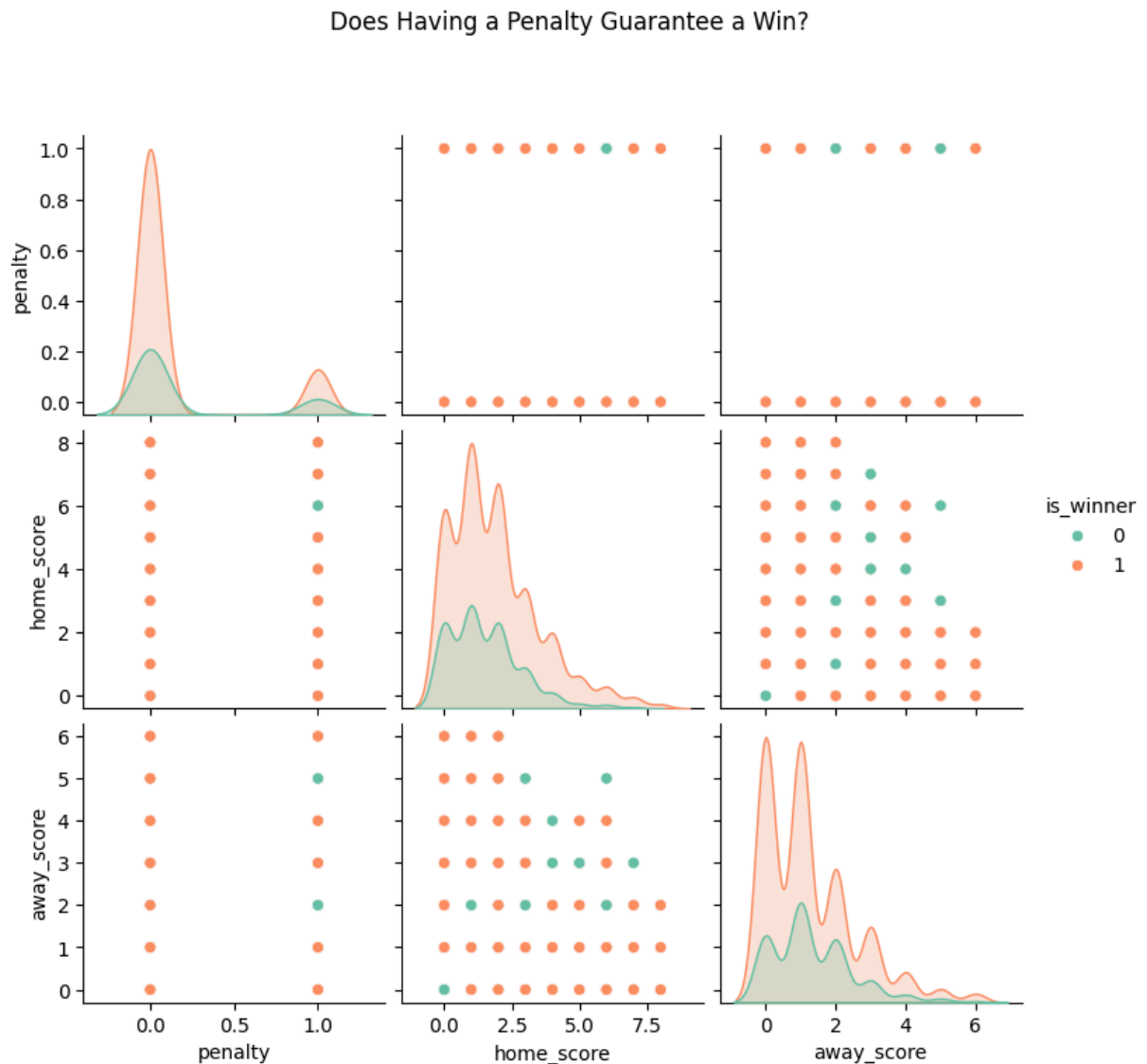
Pair Plot:

- Matches in non-neutral locations (x-axis neutral = 0) strongly favor home teams, suggesting a home-field advantage.

- Diagonal plot(KDEs) shows distribution of each variable.
- Off-diagonal scatter plot shows pairwise between the variables, in this situations the location of match, home_score, away_score.

```
In [131... columns_to_plot_pen = ['penalty', 'is_winner', 'home_score', 'away_score']

# Creating the pairplot
sns.pairplot(no_duplicates_final_with_pen[columns_to_plot_pen], hue='is_winner', di
plt.suptitle('Does Having a Penalty Guarantee a Win?', y=1.02)
plt.subplots_adjust(top=0.9) # Adjusting the layout
plt.show()
```



Pair plot:

- Distribution of having penalty kicks may not correlate with the team winning significantly.

Documentation

Key Findings

Home Score Distribution: The histogram for home_score shows the frequency distribution of home team scores. This helps to understand the range and concentration of home scores across all matches. The distribution may suggest that home teams tend to score within a specific range, providing insight into the average performance of home teams.

Home vs Away Score: The boxplot comparing home_score and away_score illustrates the differences in score distributions between the two teams. Notably, home teams appear to have a wider spread in their scores, while away teams have more concentrated score distributions. This indicates that home teams may experience more variability in their performances compared to away teams, possibly reflecting the psychological or environmental advantages of playing at home.

Penalty Frequency: The histogram for penalty shows how often penalties occur in matches. This is useful for understanding the prevalence of penalties in the dataset and assessing whether they are relatively rare or common occurrences during the games.

Penalty Distribution via KDE Plot: The KDE plot for penalty reveals the probability distribution of penalties, further emphasizing the concentration of matches with no penalties. This visualization helps in understanding whether the dataset is biased toward matches without penalties.

Logistic Regression Analysis: The regression plot showing the relationship between home_win and penalty reveals the trend in the probability of home wins based on whether a penalty occurred. The regression line suggests that the presence of penalties does not strongly correlate with home teams winning, implying that penalties may not be a significant factor in determining the match outcome.

Correlation Heatmap: The heatmap highlights correlations between penalty, is_winner, and home_win. It provides a quick overview of the relationships, showing that there is little to no strong correlation between penalties and winning outcomes. The lack of significant correlation between these variables supports the finding that penalties might not influence the likelihood of a team winning.

Proportion of Home Wins by Venue Type (Grouped Bar Plot): The grouped bar plot comparing neutral vs non-neutral venues indicates that home teams have a stronger likelihood of winning in non-neutral locations. This suggests a clear home-field advantage in matches played at familiar venues, aligning with common knowledge that home teams typically perform better.

Proportion of Wins in Penalty vs No-Penalty Scenarios (Grouped Bar Plot): The grouped bar plot for penalty vs no-penalty matches suggests a slight trend toward higher home wins in penalty situations, though the difference is not stark. This reinforces the idea that while penalties may influence match dynamics, they do not guarantee a home win.

Impact of Venue Type and Penalties on Match Outcome (Pie Charts): Pie charts show the proportion of wins and losses in neutral vs non-neutral matches and with vs without penalties. They highlight that home teams have a higher proportion of wins in non-neutral venues, while penalties appear to have a smaller influence on the final outcome.

Pair Plot Insights: The pair plot reveals that matches in non-neutral locations (with neutral = 0) strongly favor home teams. Additionally, the pairwise relationships show no strong correlations between having a penalty and winning the match. The diagonal KDE plots and off-diagonal scatter plots reinforce the finding that while match location may affect home team victories, penalties do not appear to significantly influence the outcomes.

Connection to Research Questions:

Does the location of the match correlate with home team winning? Yes, the visualizations suggest a correlation. The pair plot and grouped bar plot comparing neutral vs non-neutral matches clearly indicate that home teams are more likely to win when the match is not played in a neutral venue. This supports the hypothesis that location plays a role in determining the likelihood of a home team victory, consistent with the well-established home-field advantage in sports.

Does having a penalty guarantee a win?

No, the visualizations suggest that penalties do not guarantee a win. The regression plot, heatmap, and pie chart do not show a significant relationship between having a penalty and winning the match. While there is a slight trend toward home team victories in penalty situations, it is not strong enough to be considered a definitive factor. The pair plot further supports the lack of a clear correlation between penalties and winning, indicating that penalties are not a strong predictor of match outcomes.

Conclusion

This study analyzed key trends in international football, focusing on the influence of match location and penalties on outcomes. Our findings highlight the significant impact of venue type on match results, reaffirming the home-field advantage theory. Matches played at non-neutral venues show a clear trend of higher home team victories, as evidenced by the grouped bar plots and pair plot analyses. This correlation underscores the psychological and logistical benefits associated with playing at home, such as crowd support and familiarity with the environment.

In contrast, penalties were found to have minimal influence on determining match outcomes. Logistic regression analysis, correlation heatmaps, and grouped bar plots all indicate that while penalties may alter the dynamics of a match, they do not significantly increase the likelihood of a win. The slight trend toward home team victories in penalty

scenarios is not statistically compelling, suggesting that penalties alone are insufficient to guarantee success.

These findings provide valuable insights into the factors shaping international football matches. The strong correlation between venue type and match outcomes emphasizes the importance of strategic planning for teams playing away from home. Meanwhile, the limited impact of penalties challenges the perception of their decisive role in influencing match results. Future research could delve deeper into other match dynamics, such as player performance under pressure or the influence of coaching strategies, to further uncover the complexities of the game.

References

- <https://www.kaggle.com/datasets/martj42/international-football-results-from-1872-to-2017/data>
- <https://creativecommons.org/publicdomain/zero/1.0/>